

ANALYSIS OF CRITICAL DYNAMICS IN SIMULATED BRAIN NETWORKS AND ITS APPLICATION TO ALZHEIMER'S DISEASE

by

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Abstract

The critical brain hypothesis posits that large-scale neural networks self-organize near a continuous phase transition between ordered and chaotic dynamics, a regime hypothesized to optimize information transmission and dynamic range. In this study, we investigate the breakdown of these critical dynamics in Alzheimer’s Disease (AD) using a computational generative modeling framework. We simulated the Greenberg-Hastings cellular automaton on subject-specific structural connectomes derived from Diffusion Tensor Imaging (DTI) in a cohort of 88 participants, comprising AD, Mild Cognitive Impairment (MCI), and Subjective Cognitive Impairment (SCI) groups. We derive a theoretical framework rooted in statistical physics to characterize the network’s phase space, defining criticality through the divergence of susceptibility metrics—specifically the peak of the second-largest cluster size ($S_{max}^{(2)}$) and the maximal variance of network activity (σ_{act}). We demonstrate that simulated functional connectivity (FC) achieves maximal similarity to empirical resting-state fMRI networks precisely at this structurally defined critical threshold (T_c), reinforcing the hypothesis that functional topology emerges from structural criticality. Notably, this dynamical regime appears compromised in AD patients, who exhibit diminished susceptibility peaks and a "flattened" phase transition indicative of an extended Griffiths Phase. To quantify this deviation, we introduce a novel physics-informed metric, the Distance to Criticality (DTC), which measures the separation between a subject’s optimal operating point and their topological phase transition. Our preliminary classification analysis suggests that DTC may capture pathological features distinct from standard static correlation metrics, particularly in severe disease states. These findings imply that Alzheimer’s disease may drive a systemic shift of the brain’s dynamical working point toward a subcritical, disordered phase. While promising, further investigation is required to validate the robustness of this theoretical framework and to determine whether the "distance" to criticality can serve as a reliable clinical biomarker for neurodegenerative decline.

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Chapter 1

Introduction

The concept of criticality refers to a unique state of a complex system poised at the boundary between two distinct regimes of behavior: an ordered, subcritical phase where perturbations rapidly die out, and a disordered, supercritical phase where perturbations amplify uncontrollably [3, 4]. In the language of statistical mechanics, this boundary represents a second-order phase transition, characterized by symmetry breaking and the emergence of macroscopic complexity from microscopic interactions.

This phenomenon of emergence is central to understanding how simple local rules can give rise to sophisticated global behaviors without a central controller [5]. In systems like the Ising model, for instance, individual atomic spins follow a simple binary rule of aligning with their nearest neighbors. Yet, near the critical temperature (Curie point), these local interactions propagate across the entire lattice, leading to the sudden emergence of long-range correlations and global magnetization [6, 7]. Similarly, in biological systems, emergent behavior is observed when the collective dynamics of the system—such as the synchronized flight of starling flocks or the formation of gene regulatory networks—cannot be predicted solely by the properties of the individual components [2, 8]. This universality suggests that criticality provides a general mechanism for systems to access a vast repertoire of functional states using only local interactions [9].

The "critical-brain hypothesis" posits that neural networks are evolutionarily tuned to operate near this critical point to optimize information processing [1, 3]. In the mammalian cortex, this manifests as a delicate balance between neuronal excitation and inhibition, preventing both quiescence (silence) and saturation (seizures). This tuning is not accidental but is thought to be achieved via mechanisms of self-organized criticality (SOC), where synaptic plasticity rules autonomously adjust connection strengths to maintain the system at the phase transition without the need for external parameter fine-tuning [4, 10]. Here, the "simple rule" of homeostatic plasticity—adjusting synaptic weights to maintain a target firing rate—allows the macroscopic network to self-assemble into a regime of maximal sensitivity [10].

Systems operating at criticality exhibit specific, universal hallmarks. Foremost

among these are scale-invariance and power-law distributions in spatiotemporal activity patterns [3]. Functionally, the critical regime is associated with the maximization of dynamic range, information transmission, and storage capacity. At this point, the susceptibility of the system diverges, granting the network high sensitivity to external stimuli while maintaining stability [11]. Furthermore, criticality facilitates the emergence of "small-world" network topologies, optimizing the trade-off between local segregation (specialized processing) and global integration [1].

Historically, evidence for neural criticality was first identified in the temporal correlations of EEG signals and subsequently in the "neuronal avalanches" observed in local field potentials (nLFPs) of cortical slices [1, 3]. These avalanches were found to follow power-law size distributions with exponents characteristic of a branching process near unity [12]. More recently, computational studies have demonstrated that large-scale resting-state networks (RSNs) observed in fMRI emerge naturally when dynamic models are implemented on the structural human connectome at the critical point [12, 13].

1.1 Literature Survey

The investigation of brain dynamics through the lens of statistical physics began with the analysis of long-range temporal correlations (LRTC) in electrophysiological data. Early work established that spontaneous neural fluctuations obey power-law scaling, a signature of critical states [3]. Seminal work by Beggs and Plenz provided experimental verification of this hypothesis in vitro, demonstrating that the propagation of spontaneous activity in cortical networks occurs as neuronal avalanches. The size (S) and duration (D) of these avalanches follow power-law distributions $P(S) \sim S^{-\tau}$ and $P(D) \sim D^{-\beta}$, satisfying the scaling relationship $\beta = \tau - 1$ predicted by the theory of critical branching processes [3, 12].

Expanding to the macroscopic scale, Haimovici et al. bridged the gap between structural and functional connectivity. By simulating critical dynamics on a human connectome derived from diffusion MRI (dMRI), they showed that the resulting simulated Resting State Networks (RSN) matrices exhibited maximal similarity to empirical fMRI data only when the system operated at the critical phase transition [12]. This strongly suggests that the structural scaffold of the brain supports a dynamic regime specifically tuned to criticality to facilitate the diverse repertoire of functional states observed in the healthy brain.

To model these dynamics computationally, the Greenberg-Hastings (GH) model serves as a robust framework. Originally conceptualized in the context of excitable media, the GH model is a cellular automaton where nodes transition cyclically through three states: Quiescent (susceptible), Excited (active), and Refractory (recovering) [13]. Unlike continuous oscillator models, the discrete nature of the GH model captures the binary,

pulse-like nature of neuronal firing while remaining computationally efficient for large-scale network simulations.

Recent analytical derivations have characterized the phase space of the GH model on complex networks. Almeida et al. demonstrated that the spontaneous activation probability (r_1) acts effectively as an external field conjugate to the order parameter. As $r_1 \rightarrow 0$, the system exhibits clear finite-size scaling in susceptibility, a hallmark of continuous phase transitions [11]. Furthermore, extensions of the model incorporating excitatory and inhibitory (E/I) balance reveal a rich phase diagram, where the transition can shift from first-order (discontinuous) to second-order (continuous) depending on the fraction of inhibitory nodes [14].

A crucial refinement in whole-brain modeling is the incorporation of homeostatic plasticity. Rocha et al. introduced a normalization mechanism for the structural connectivity matrix that equalizes the effective in-degree of nodes. This adjustment, biologically analogous to synaptic scaling, reduces inter-subject variability and causes macroscopic observables to collapse onto universal curves, thereby stabilizing the critical point across different structural topologies [13].

Another approach for the problem of getting an activation history of fixed duration is shown by Enzo *et.al* [15]. The main idea is to get rid of the dynamic mechanism and hence the simulation; instead, utilizing a novel approach of deconvolution of BOLD signals to arrive at a point process. Their work has been valuable for us since we have also attacked the same problem. The approach of not using a dynamic mechanism was also used to verify the results coming from the simulation.

This framework has proven powerful in clinical neuroscience. Studies on stroke recovery have shown that lesions force the brain into a sub-critical, low-entropy state with diminished functional connectivity. Remarkably, behavioral recovery correlates with the "re-tuning" of the system back toward criticality, driven by white-matter remodeling [16]. This establishes criticality not just as a theoretical curiosity, but as a quantifiable biomarker of brain health and plasticity.

Building upon these foundations, this project investigates the disruption of critical dynamics in Alzheimer's Disease (AD). We hypothesize that the progression of neurodegeneration—from Subjective Cognitive Impairment (SCI) through Mild Cognitive Impairment (MCI) to AD—systematically degrades the brain's ability to maintain a critical state [4]. Specifically, we postulate that structural damage leads to a "duller" phase transition, characterized by a reduction in the peak magnitude of the second-largest cluster (SLC) and susceptibility. By applying the GH model to patient cohorts, we aim to quantify these deviations, validating criticality-based metrics as potential non-invasive biomarkers for the early detection and tracking of cognitive decline.

At this critical point (often denoted by a branching ratio $\sigma \approx 1$), the system exhibits:

- Maximal dynamic range. [3]
- Scale-free distributions of neuronal avalanches (power-law scaling). [3]
- Optimized information transmission and storage [17].

This project evaluates these properties by modeling flow on structural connectomes derived from MRI data, comparing healthy control groups against cohorts with Subjective Cognitive Impairment (SCI), Mild Cognitive Impairment (MCI), and Alzheimer's Disease (AD).

Chapter 2

Materials and Methods

2.1 The Greenberg-Hastings (GH) Model

We simulate neural dynamics using a 3-state Greenberg-Hastings (GH) cellular automaton model. Each node i in the network (representing a brain region) can be in one of three discrete states at time t :

- $S_i(t) = 0$: **Refractory (R)**, or temporarily inactive.
- $S_i(t) = 1$: **Excited (E)**, or active/firing.
- $S_i(t) = 2$: **Quiescent (Q)**, or susceptible to activation.

The state of all N nodes is updated in parallel at each discrete time step t according to the following transition rules, governed by three main parameters: the activation threshold (T), the spontaneous activation rate (r_1), and the recovery rate (r_2).

1. **Excited $\rightarrow$ Refractory (E $\rightarrow$ R)**: Any node that is excited at time t deterministically becomes refractory at $t + 1$.

$$P(S_i(t+1) = 0 \mid S_i(t) = 1) = 1$$

2. **Refractory $\rightarrow$ Quiescent (R $\rightarrow$ Q)**: A node in the refractory state at time t transitions to the quiescent state at $t + 1$ with a probability r_2 .

$$P(S_i(t+1) = 2 \mid S_i(t) = 0) = r_2$$

3. **Quiescent $\rightarrow$ Excited (Q $\rightarrow$ E)**: A quiescent node i can become excited in two ways:

- (a) **Spontaneously**: With a small probability r_1 .

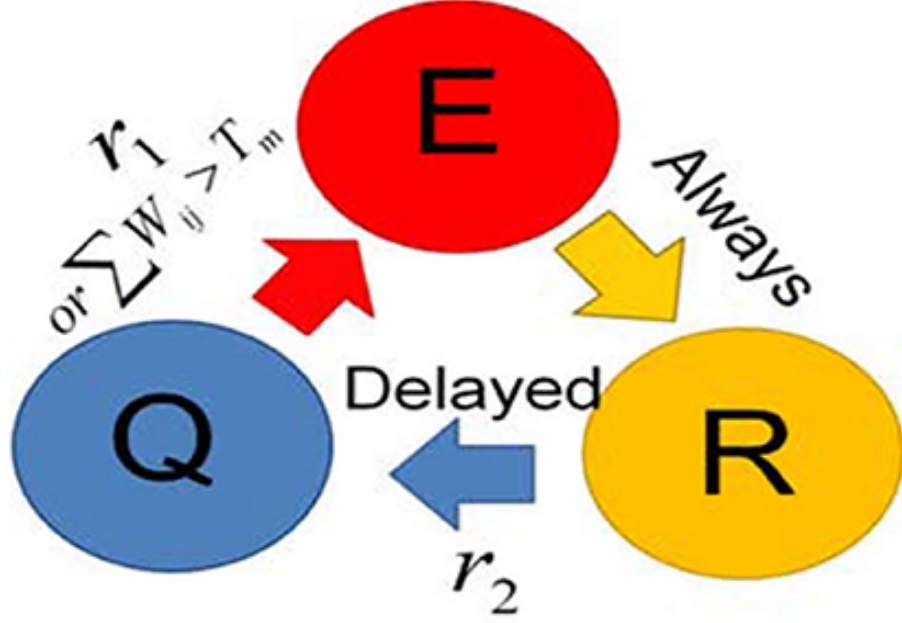


Figure 2.1: Greenberg Hasting transition rules. This photo was taken from [1]

- (b) **Driven:** If the total weighted input from its excited neighbors j (where $S_j(t) = 1$) exceeds the activation threshold T .

In conformity with the paradigm, structural connectivity matrix is row-wise normalized [2] [18]. The normalization process can be also thought as a adaptive threshold mechanism in which instead of using a global threshold parameter for each node with varying degree, threshold is scaled with the nodal degrees, mimicking the neuroplasticity mechanism in the brain. Let W' be the normalized structural connectivity matrix. It is computed from the original structural connectivity matrix, W as follows:

$$W'_{ij} = \frac{W_{ij}}{\sum_{i=1}^N W_{ij}}$$

The total input to node i is $I_i(t) = \sum_{j=1}^N W'_{ij} \cdot \delta(S_j(t) = 1)$, where $\delta(\cdot)$ is the Kronecker delta. The node becomes active if $I_i(t) > T$. The full transition probability is:

$$P(S_i(t+1) = 1 \mid S_i(t) = 2) = 1 - (1 - r_1)[1 - \Theta(I_i(t) - T)]$$

where $\Theta(\cdot)$ is the Heaviside step function. In our study, we fix r_1 to a small value ($r_1 = 0.01$) and explore the 2D parameter space defined by the control parameter T (threshold) and r_2 (recovery rate).

2.2 Simulation Framework

The simulation framework is designed using an Object-Oriented Programming (OOP) architecture in MATLAB to ensure modularity and reproducibility. The complete source code for this framework is available at https://github.com/borankilic/critical_brain. The core of the framework relies on the separation of static network topology from dynamic state evolution, implemented through two primary classes: `Network` and `GreenbergHastingsModel`. The `Network` class serves as the fundamental data structure for handling structural connectivity. It encapsulates the weighted adjacency matrix (W), region of interest (ROI) labels, and atlas parcellations. This class is responsible for preprocessing the physical substrate of the simulation, offering methods for graph-theoretic operations such as row-normalization of the connectivity matrix, which is essential for standardizing input currents across nodes with heterogeneous degrees.

To standardize the simulation interface, we implemented an abstract base class named `DynamicsModel`. This superclass defines the essential properties required for any time-stepping simulation, such as the reference to a `Network` object, the number of simulation steps (T_{max}), and the state history matrices. The specific simulation logic is implemented in the `GreenbergHastingsModel` class, which inherits from `DynamicsModel`. This subclass manages the specific parameters of the automaton:

- **State Properties:** It maintains the current state of all N nodes (Quiescent, Excited, Refractory) and stores the full history of binary activation patterns.
- **Parameter Control:** The class exposes properties for the activation threshold (T), spontaneous activation rate (r_1), and recovery probability (r_2), allowing for programmatic parameter sweeps.
- **Integrated Cluster Detection:** Unlike standard time-stepping solvers, the `run` method of this class integrates graph traversal algorithms (connected components) directly into the simulation loop via the `cluster_iterate` function. This allows for the real-time calculation of criticality metrics, such as the size of the largest and second-largest active clusters, at every time step.

2.3 Dataset

The brain is modeled as a complex network defined by two complementary modalities: Structural Connectivity (SC) and Functional Connectivity (FC).

Structural Connectivity (SC) is derived from Diffusion Tensor Imaging (DTI) data. In this framework, the brain is parcellated into distinct Regions of Interest (ROIs) which serve as the network nodes. The connections (edges) between these nodes represent

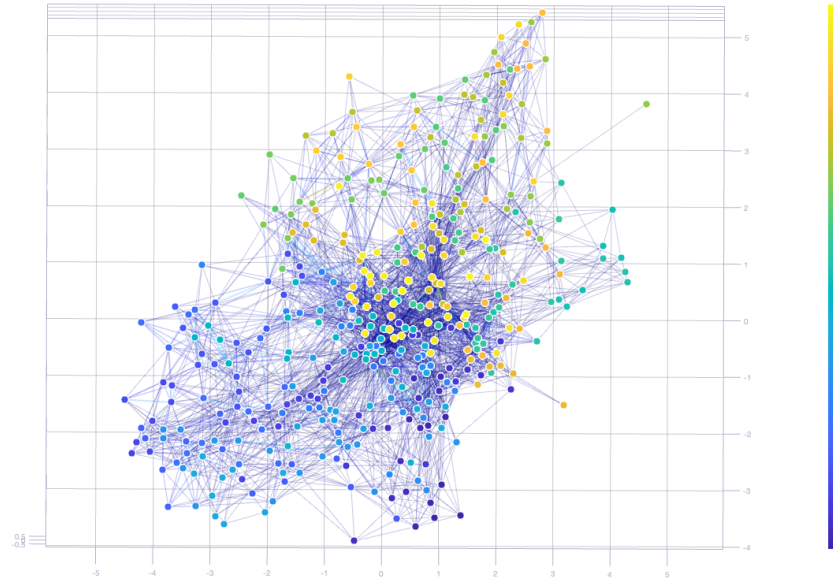


Figure 2.2: Structural network of a healthy patient is visualised inside Network class.

the physical white-matter tracts (axons) identified via tractography. The weight of these connections typically signifies the fiber density or streamline count, providing the physical substrate that constrains information propagation.

Functional Connectivity (FC) is obtained from resting-state functional MRI (rs-fMRI) data. Unlike the static physical wiring of SC, FC represents the temporal correlations (typically Pearson correlation) between the Blood-Oxygen-Level-Dependent (BOLD) signals of different ROIs. It captures the dynamic synchronization of neural activity that emerges from the underlying structural topology.

Patient Data

We use a cohort of structural (DTI) and functional (rs-fMRI) network data acquired from our lab's (VAVLAB) collaboration with CAPA hospital. The cohort consists of three groups:

- **SCI:** Subjective Cognitive Impairment (N= 23)
- **MCI:** Mild Cognitive Impairment (N= 43)
- **AD:** Alzheimer's Disease (N= 14)

All data was processed using our in-house preprocessing and post-processing pipeline to generate SC and FC matrices for each participant.¹

¹<https://vavlab.bogazici.edu.tr/brainet-structural-and-functional-brain-network-analysis>

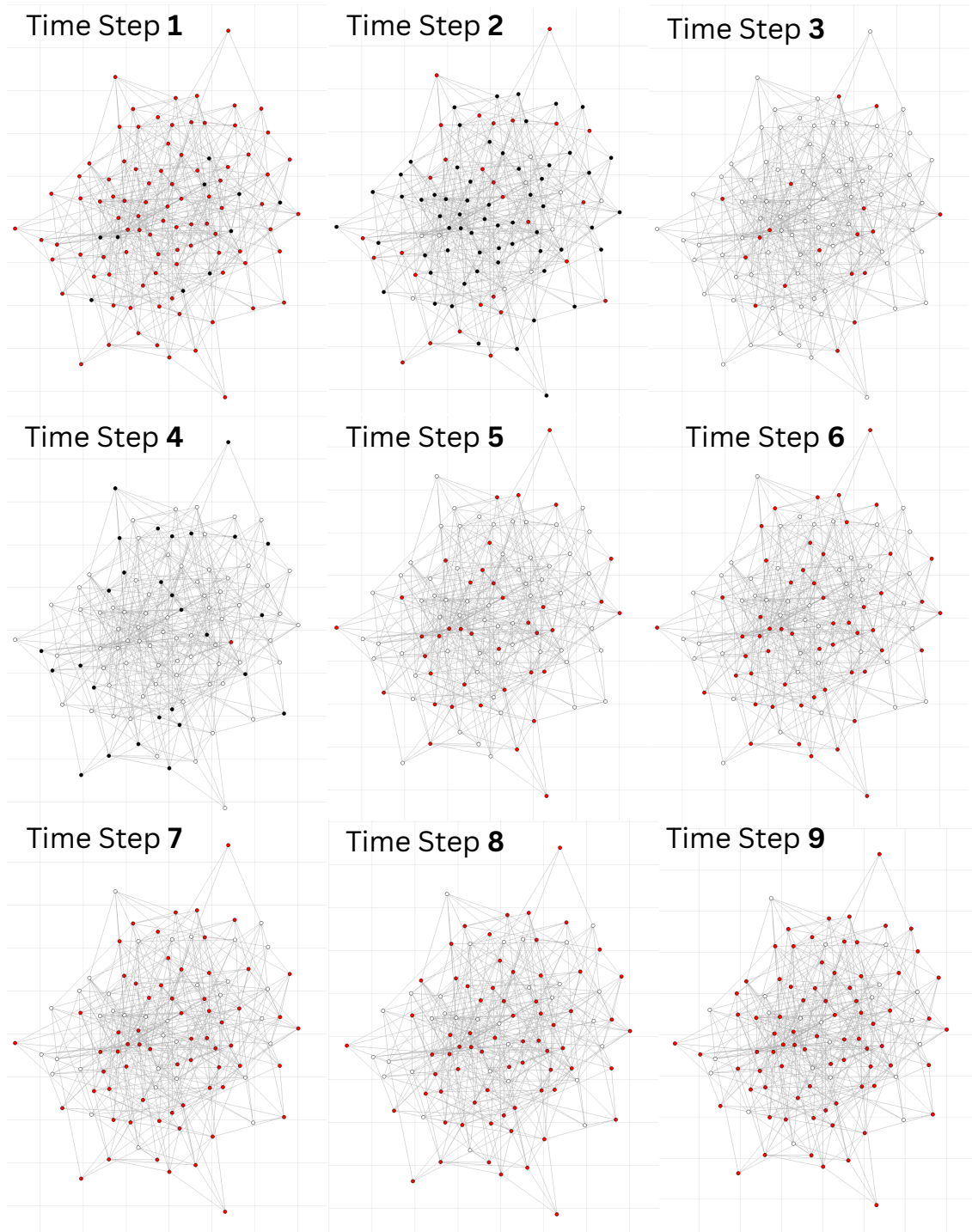


Figure 2.3: First eight time steps of the GH simulation on a random small-world network ($N = 200$, $k = 6$). Parameters: 10% initial excitation, $r_1 = 0.5$, $r_2 = 0.2$, and threshold = 0.2. Red represent quiescent, white excited and black refractory nodes.

Chapter 3

Theoretical Fundamentals

3.1 Mean Field Theory of State Populations in Dynamic GH model

Let $x_i(t)$ represent the state of i -th node at time step t where

$$x_i(t) = \begin{cases} 0 & , \text{refractory} \\ 1 & , \text{excited} \\ 2 & , \text{quiescent} \end{cases} \quad (3.1)$$

Update dynamics of the GH model are given as in the table below.

Table 3.1: State Transition Rules for $x_i(t)$

Current State $x_i(t)$	Next State $x_i(t+1)$	Condition/Probability	Notes
0	2	with probability $\tilde{r}_2$	Spontaneous activation.
1	0	with probability 1	Guaranteed deactivation.
2	1	with probability $\tilde{r}_1$	External excitation
2	1	if $\sum_{j \in \text{Neighbors}} x_j(t) > T$	Neighbor influence transition, where T is the threshold.

Let us define the probabilities any node being in one of the 3 GH states as:

$$r_i(t) = \Pr \{x_i(t) = 0\} \quad (3.2)$$

$$e_i(t) = \Pr \{x_i(t) = 1\} \quad (3.3)$$

$$q_i(t) = \Pr \{x_i(t) = 2\} \quad (3.4)$$

$$(3.5)$$

The simulation in general is sthochastic due to external excitation probability $\tilde{r}_1$ and random initialization of neurons. It can also be considered undeterministics if model is started from a known congiguration and $r_1 = 0$. However, independent of whether the simulations are sthochastic or deterministics, we consider $x_i(t)$ as a random variable in the state space of all possible configurations.

Using transition dynamic of GH model:

$$r_i(t+1) = e_i(t) + (1 - \tilde{r}_2) \cdot r_i(t) \quad (3.6)$$

$$q_i(t+1) = \tilde{r}_2 \cdot r_i(t) + (1 - p_i(t)) \cdot q_i(t) \quad (3.7)$$

$$e_i(t+1) = (\tilde{r}_1 + p_i(t)) \cdot q_i(t) \quad (3.8)$$

where

$$p_i(t) = \Pr \left\{ \sum_{j \in \mathcal{N}(i)} W_{ij} \delta_{1, x_j(t)} = W_i X_i(t) > T \right\} = \Pr \{ \Theta(W_i X(t) - T) = 1 \} \quad (3.9)$$

where $\Theta(\cdot)$ is the Heaviside function and $X(t) \in \{0, 1\}$ excited state vector. As $t \rightarrow \infty$ or practically for a large enough time, system will settle to its steady state value.

$$\{e, r, q\}_i(t) = \{e, r, q\}_i(t+1) = \{e^*, r^*, q^*\}_i \quad \text{as } \rightarrow \infty \quad (3.10)$$

and also $p_i(t) \rightarrow p_i^*$. Then equations 3.6 becomes:

$$r_i^* = e_i^* + (1 - r_2) \cdot r_i^* \Rightarrow r_i^* = \frac{e_i^*}{r_2} \quad (3.11)$$

Using $q_i^* + e_i^* + r_i^* = 1$, we get

$$q_i^* = 1 - e_i^* - r_i^* = 1 - e_i^* \left(1 + \frac{1}{\tilde{r}_2} \right). \quad (3.12)$$

Finally, rewiriting 3.8

$$e_i^* = (\tilde{r}_1 + p_i^*) \cdot \left(1 - e_i^* \left(1 + \frac{1}{\tilde{r}_2} \right) \right) \Rightarrow e_i^* = \frac{\tilde{r}_1 + p_i^*}{1 + (\tilde{r}_1 + p_i^*) (1 + 1/\tilde{r}_2)}. \quad (3.13)$$

In a homegeneous operating point $e_i^* = e^*$, this would become a scalar fixed-point equation of the form

$$e^* = \frac{\tilde{r}_1 + F(e^*)}{1 + (\tilde{r}_1 + F(e^*)) (1 + 1/\tilde{r}_2)}. \quad (3.14)$$

where $F(e^*)$ is the approximation to $\Pr\{M_i > T\}$ under e^* . Unfortunately, structural connections in brain are sparse and highly non-homogeneous. Thus, one has the N-

dimensional fixed-point problem which dictates as

$$\mathbf{e}^* = \frac{\tilde{r}_1 + F(\mathbf{e}^*)}{1 + (\tilde{r}_1 + F(\mathbf{e}^*)) (1 + 1/\tilde{r}_2)}. \quad (3.15)$$

where $\mathbf{e}^* = [e_1^* \ e_2^* \dots e_N^*]$. This can be solved using numerical iteration techniques given we start from a close enough point to the solution so that it converges.

3.2 Perturbations Around Steady State

Now, we study small perturbation around (e^*, r^*) to study stability of steady-state solutions. In another words, we are trying to answer whether steady-state attract or repel perturbations. In the midst of this tug of war between chaotic and ordered behaviour lies the “critical point” where perturbations self-consistently oscillate around steady-state.

Firstly, we define derivatives

$$A_{ij} = \left. \frac{\partial p_i(\mathbf{e})}{\partial e_j} \right|_{\mathbf{e}=\mathbf{e}^*} \quad (3.16)$$

which is representative of the dynamic response of node j to node i . It measures how much a small change in the probability of node j being excited affects the excitation of node i which can be referred to as “dynamical connectivity”.

To be able to evaluate this, we write 3.8 as an expectation value over all possible binary states

$$\begin{aligned} p_i(e) &= \mathbb{E} [\Theta(M_i > T)] = \sum_{\mathbf{x} \in \{0,1\}^N} \Theta \left(\sum_j W_{ij} x_j - T \right) \prod_j e_j^{x_j} (1 - e_j)^{1-x_j} \\ &= \sum_{\mathbf{x} \in \{0,1\}^{|\mathcal{N}(i)|}} \Theta \left(\sum_{j \in \mathcal{N}(i)} W_{ij} x_j - T \right) \prod_{j \in \mathcal{N}(i)} e_j^{x_j} (1 - e_j)^{1-x_j}. \end{aligned} \quad (3.17)$$

Equation above reduces to sum / multiplication over neighbouring nodes only since the first term inside the summation is always 0 or 1 during summation over second term which makes the second term over non-neighbors sum to 1, because it's the full probability of all binary configurations of unrelated Bernoulli variables.

$$\begin{aligned} \frac{\partial p_i}{\partial e_\ell} &= \sum_{\mathbf{x} \in \{0,1\}^{|\mathcal{N}(i)|}} \Theta \left(\sum_{j \in \mathcal{N}(i)} W_{ij} x_j - T \right) \frac{\partial}{\partial e_\ell} \left[e_\ell^{x_\ell} (1 - e_\ell)^{1-x_\ell} \right] \prod_{\substack{j \neq \ell \\ j \in \mathcal{N}(i)}} e_j^{x_j} (1 - e_j)^{1-x_j} \\ &= \sum_{\mathbf{x} \in \{0,1\}^{|\mathcal{N}(i)|}} \Theta \left(\sum_{j \in \mathcal{N}(i)} W_{ij} x_j - T \right) \left(\frac{x_\ell}{e_\ell} - \frac{1 - x_\ell}{1 - e_\ell} \right) \prod_{j \in \mathcal{N}(i)} e_j^{x_j} (1 - e_j)^{1-x_j} \end{aligned} \quad (3.18)$$

Notice one can also write the equations above, in terms of conditional probabilities, as

$$p_i = e_\ell \Pr(M_i > T \mid X_\ell = 1) + (1 - e_\ell) \Pr(M_i > T \mid X_\ell = 0). \quad (3.19)$$

and

$$\frac{\partial p_i}{\partial e_\ell} = \Pr(M_i > T \mid X_\ell = 1) - \Pr(M_i > T \mid X_\ell = 0). \quad (3.20)$$

which, intuitively, tells us about how much greater the spontaneous excitation probability of node- i when node- ℓ is excited compared to when it's not.

Now we define the first-order perturbations of state state probabilities $(\delta e_i, \delta r_i)$ from steady-state values (e_i^*, r_i^*) of node- i at time t as

$$\delta e_i(t) = e_i(t) - e_i^*, \quad \delta r_i(t) = r_i(t) - r_i^*, \quad \delta q_i(t) = -\delta e_i(t) - \delta r_i(t) \quad (3.21)$$

Let us linearize the nonlinear system around the fixed point by taking the first order Taylor-expansion of the update map. Full nonlinear update map $F(\cdot)$ is

$$\begin{pmatrix} e_i(t+1) \\ r_i(t+1) \end{pmatrix} = F \begin{pmatrix} e_i(t) \\ r_i(t) \end{pmatrix} = \begin{pmatrix} (\tilde{r}_1 + p_i(\mathbf{e})) \cdot (1 - e_i(t) - r_i(t)) \\ e_i(t) + (1 - \tilde{r}_2) r_i(t) \end{pmatrix}. \quad (3.22)$$

Expanding $F \begin{pmatrix} e_i(t) \\ r_i(t) \end{pmatrix}$ to the first order

$$F \begin{pmatrix} e_i^* + \delta e_i(t) \\ r_i^* + \delta r_i(t) \end{pmatrix} = F(e^*, r^*) + J \begin{pmatrix} \delta e \\ \delta r \end{pmatrix} + O(\|\delta\|^2) \quad (3.23)$$

Either treating the perturbations as partial differentials (taking differential of equations 3.6 and 3.8) or following the more rigorous proof at B, one arrives at

$$\delta e_i(t+1) = q_i^* \cdot \sum_j A_{ij} \delta e_j - \left(\tilde{r}_1 + p_i(\mathbf{e}^*) \right) \cdot \left(\delta e_i(t) + \delta r_i(t) \right) \quad (3.24)$$

$$\delta r_i(t+1) = \delta e_i(t) + (1 - \tilde{r}_2) \delta r_i(t) \quad (3.25)$$

In the vector form

$$\delta \mathbf{e}(t+1) = \mathbf{Q}^* \mathbf{A} \cdot \delta \mathbf{e}(t) - \hat{\mathbf{P}} \cdot \delta \mathbf{e}(t) - \hat{\mathbf{P}} \cdot \delta \mathbf{r}(t) \quad (3.26)$$

$$\delta \mathbf{r}(t+1) = \delta \mathbf{e}(t) + (1 - \tilde{r}_2) \mathbf{I} \cdot \delta \mathbf{r}(t) \quad (3.27)$$

where

$$\delta \mathbf{e}(t) = [\delta e_1(t) \delta e_2(t) \dots \delta e_N(t)]^T, \quad \delta \mathbf{r}(t) = [\delta r_1(t) \delta r_2(t) \dots \delta r_N(t)]^T$$

$$\mathbf{Q}^* = \text{diag}[q_i^*] \quad , \quad \hat{\mathbf{P}} = r_1 \mathbf{I} + \text{diag}[p_i^*(\mathbf{e})]$$

Let us define the augmented state perturbation vector

$$\Delta(t) = \begin{bmatrix} \delta \mathbf{e}(t) \\ \delta \mathbf{r}(t) \end{bmatrix} \in \mathbb{R}^{2N}. \quad (3.28)$$

Evolution of perturbations from steady state are dictated by the Jacobian matrix $\mathbf{J}$ as

$$\Delta(t+1) = \begin{bmatrix} \mathbf{Q}^* \mathbf{A} - \hat{\mathbf{P}} & -\hat{\mathbf{P}} \\ \mathbf{I} & (1 - \tilde{r}_2) \mathbf{I} \end{bmatrix} \Delta(t) = \mathbf{J} \Delta(t) \quad (3.29)$$

The linearized perturbations evolve according to

$$\Delta(t+1) = \mathbf{J} \Delta(t), \quad \Delta(t) = \mathbf{J}^t \Delta(0), \quad (3.30)$$

so the stability of the steady state is determined entirely by the eigenvalues of the Jacobian. Writing the eigendecomposition $\mathbf{J} = \mathbf{V} \Lambda \mathbf{V}^{-1}$, one obtains

$$\Delta(t) = \sum_k c_k \lambda_k^t v_k, \quad (3.31)$$

showing that each mode grows or decays with rate $|\lambda_k|^t$. Defining the spectral radius $\rho(\mathbf{J}) = \max_k |\lambda_k|$, the stability of steady-state are dependent on the eigenvalues of $\mathbf{J}$ as:

- $\rho(\mathbf{J}) < 1 \Rightarrow$ convergent / stable / subcritical
- $\rho(\mathbf{J}) > 1 \Rightarrow$ divergent / unstable / supercritical
- $\rho(\mathbf{J}) \simeq 1 \Rightarrow$ critical

Eigenvalues of $\mathbf{J}$ can be calculated from

$$\det(\mathbf{J} - \lambda \mathbf{I}) = \det(\mathbf{D} - \lambda \mathbf{I}) \cdot \det(\lambda \mathbf{I} + \hat{\mathbf{P}} - \mathbf{Q}^* \mathbf{A} + \frac{\hat{\mathbf{P}}}{\lambda + \tilde{r}_2 - 1}) = 0 \quad (3.32)$$

which reveals that first N eigenvalues are $1 - \tilde{r}_2$ and others are found by solving the polynomial by expanding the determinant in the second term.

Chapter 4

Analysis

This chapter details the experimental results obtained from simulating the GH model on human connectomes. We analyze the transition from ordered to chaotic phases and evaluate the model's ability to reproduce empirical functional connectivity (FC).

4.1 Experiment 1: Cluster Analysis and Activity Statistics

4.1.1 Mean and Standard Deviation of Activity

We swept the parameter space (T, r_2) to observe the mean network activity $\langle A \rangle$ and its temporal variability σ_A .

$$\langle A(T, r_2) \rangle = \frac{1}{\mathcal{T}} \sum_{t=1}^{\mathcal{T}} \sum_{i=1}^N a_i(t; T, r_2) \quad (4.1)$$

$$\sigma_A(T, r_2) = \sqrt{\frac{1}{\mathcal{T}-1} \sum_{t=1}^{\mathcal{T}} \left(\sum_{i=1}^N a_i(t; T, r_2) - \langle A(T, r_2) \rangle \right)^2} \quad (4.2)$$

- **Ordered Phase:** Low activity, low variance.
- **Chaotic Phase:** High activity, low variance (saturation).
- **Critical Regime:** Intermediate activity with maximized variance σ_A , indicating high susceptibility to inputs.

As discussed in [18], the standard deviation of the activity is expected to peak around the critical point while the mean activity monotonically decreases as the threshold increases. This is what we have observed with our simulations as well.

4.1.2 Largest and Second Largest Clusters

A group of structurally connected nodes that are active at a time instance is said to be a cluster at that time instance.

$$i \in C(t) \iff \left(a_i(t) = 1 \wedge \exists j \in C(t) : S_{ij} \neq 0 \right) \quad (4.3)$$

For the analysis of the criticality phenomena, the second largest cluster possesses a crucial role. It has been theorized that a dynamic mechanism of a complex network operating far from the critical point forms only small-sized second largest clusters, while the size of the largest cluster steadily declines as the activity goes from super-critical to sub-critical [12]. It is important here to note that in theory the point where the second largest cluster occurs is exactly the critical point, but as has been explained in [16], the finite size of the network causes the phenomena to occur near the critical point instead of precisely at the critical point.

For the purpose of our project we have utilized the time-averaged second-largest cluster sizes.

$$\langle C_2 \rangle = \frac{1}{T} \sum_{t=1}^T C_2(t) \quad (4.4)$$

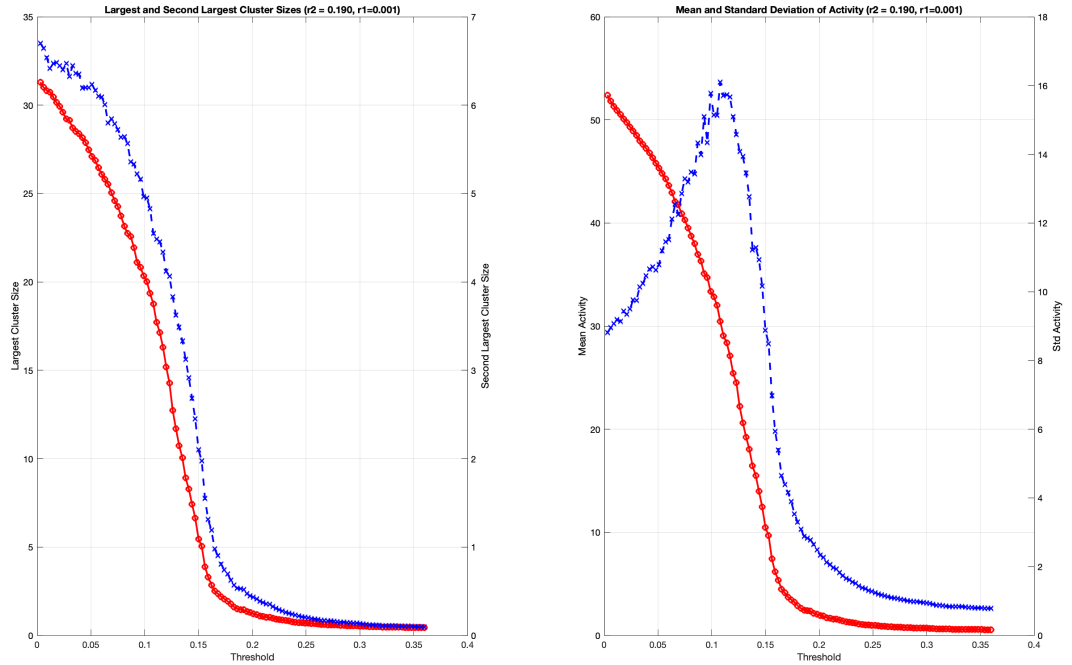


Figure 4.1: Largest (Red) and second largest (Blue) cluster sizes on the left; mean activity (Red circles) and standard deviation of activity (Blue crosses) on the left at $\mathbf{r}_2 = 0.19$.

As can be seen from **Figure 4.1**, **Figure 4.2** and **Figure 4.3**, the critical point seems to be around 0.15 for this patient. Because of the finite-size effect, the critical point

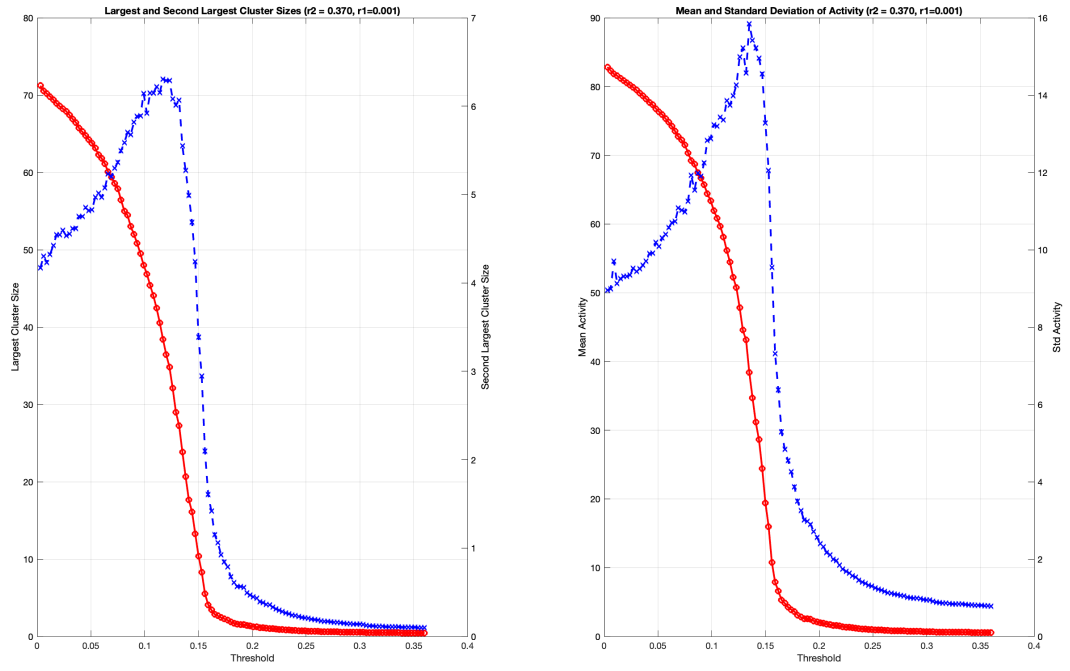


Figure 4.2: Largest (Red) and second largest (Blue) cluster sizes on the left; mean activity (Red circles) and standard deviation of activity (Blue crosses) on the left at $r_2 = \mathbf{0.37}$.

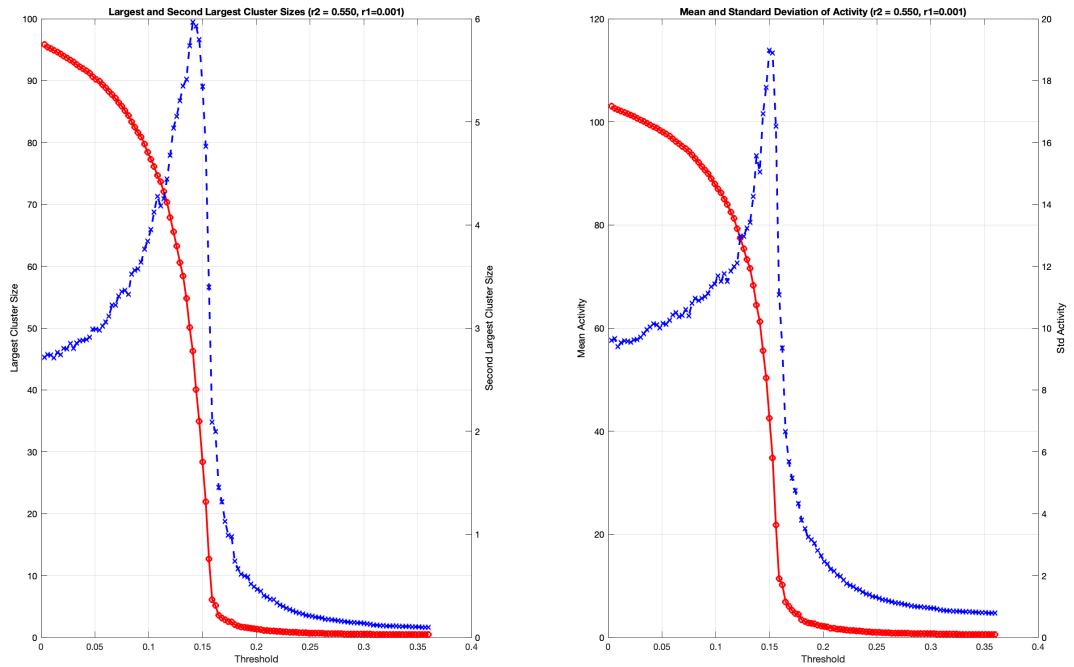


Figure 4.3: Largest (Red) and second largest (Blue) cluster sizes on the left; mean activity (Red circles) and standard deviation of activity (Blue crosses) on the left at $r_2 = \mathbf{0.55}$.

is not exactly at 0.15 in left subfigure of **Figure 4.2**, which is expected.

To fully characterize the robustness of the critical state, we extended our analysis to a 3D parameter space by varying the recovery rate (r_2) alongside the threshold (T). As illustrated in **Figure 4.4** and **Figure 4.5**, both the second-largest cluster size ($S_{max}^{(2)}$) and the standard deviation of activity (σ_{act}) exhibit a distinct "critical ridge" (yellow regions) where the system reaches maximal susceptibility¹. This ridge represents the phase transition boundary across different recovery dynamics. Crucially, we observe that the peaking behavior—the local maxima defining the critical point—becomes significantly broader and less distinct for low recovery rates ($r_2 < 0.2$)². This suggests that if the refractory period is too long (low r_2), the system struggles to sustain the complex, self-organized fluctuations necessary for criticality, effectively dampening the phase transition.

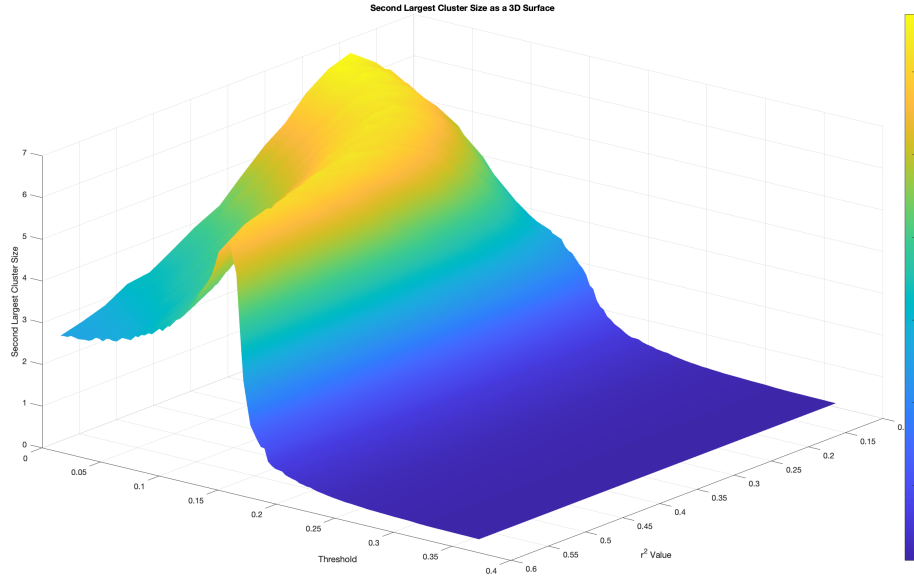


Figure 4.4: 3D surface visualization of the time-averaged second-largest cluster size ($\langle C_2 \rangle$) as a function of the activation threshold and the refractory parameter r_2 . The ridge (yellow peak) indicates the critical phase transition, which becomes less pronounced and eventually vanishes as r_2 decreases toward zero.

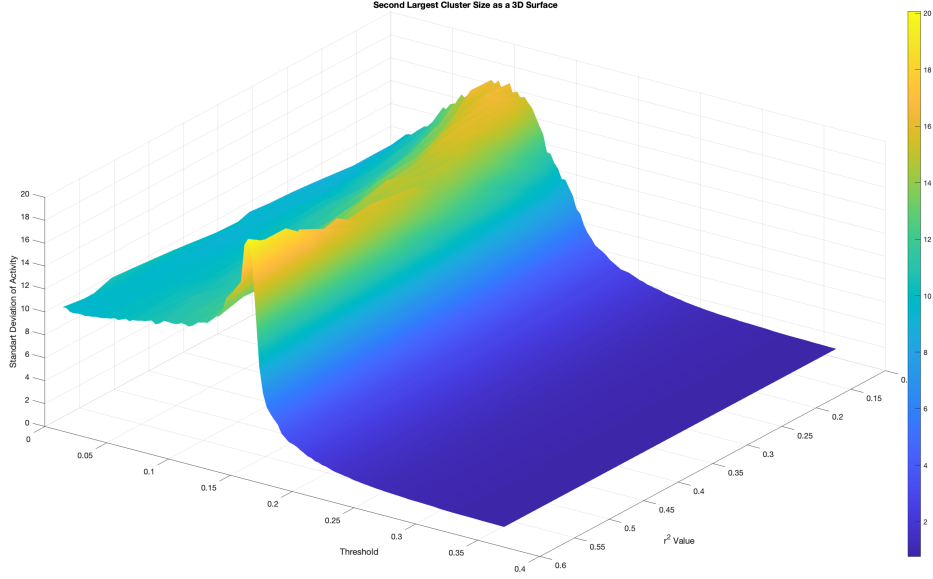


Figure 4.5: 3D surface visualization of the time-averaged standard deviation of excited population as a function of the activation threshold and the refractory parameter r_2 . The ridge (yellow peak) indicates the critical phase transition, which doesn't vanish as r_2 decreases in this case.

4.1.3 Power Law Regarding Cluster Size Distribution

In theory [19] [3], the distribution of the cluster sizes follows a power law

$$P(s) \sim s^{-\alpha} \quad s = \text{cluster size} \quad (4.5)$$

As can be seen quite clearly,

$$P(s)$$

meets the criteria for being scale-free, which is stated as:

$$P(\lambda s) = A(\lambda)P(s) \quad (4.6)$$

So in a log-log plot we expect to see a line with tangent α .

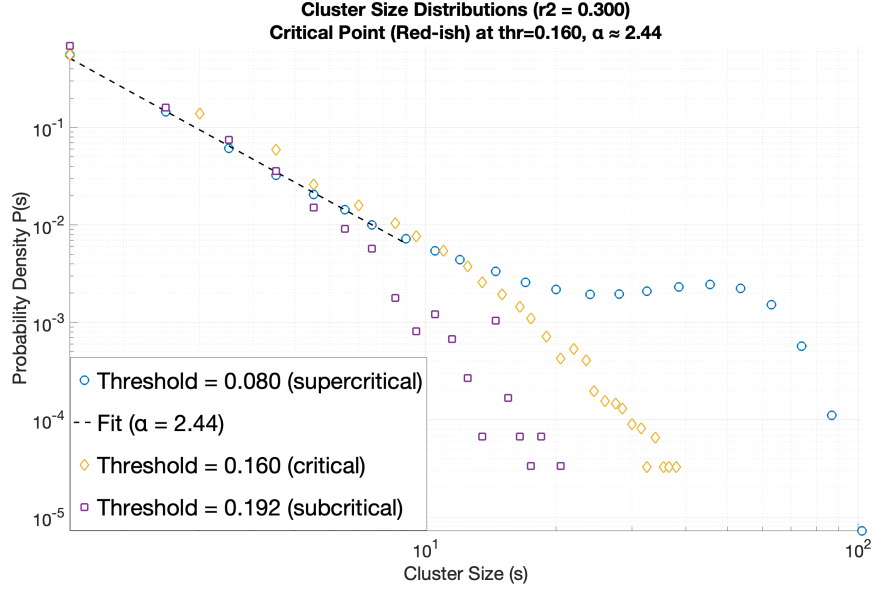


Figure 4.6: Log-log graphic showing compatibility with the power law for critical (yellow), sub-critical (purple) and super-critical (blue) cases.

As can be seen, the critical point is ≈ 0.15 which was predicted by the maximum of the second largest cluster size. The power law graph is another hallmark of criticality and the prediction of the critical point from it overlaps the prediction from the peak of second largest cluster. This further strengthens that there is a critical point in the brain.

4.2 Experiment 2: Mutual Information Analysis

To validate the information processing capacity of the network, we utilized information-theoretic measures [17, 20, 21]. The Critical Brain Hypothesis posits that neural systems operating near a second-order phase transition point maximize their dynamic range and information transmission capabilities.

Intuitively, this efficiency arises from the specific balance between order and chaos found at the critical point. In a *subcritical* (ordered) regime, neural activity dampens quickly, preventing signals from propagating to distant regions and limiting the network's ability to integrate information. Conversely, in a *supercritical* (chaotic) regime, activity spreads indiscriminately, saturating the network and drowning out specific information in noise. Only at the critical phase transition does the network maintain long-range correlations without saturation, enabling the system to integrate information from distant regions while preserving the distinctiveness of local signals [22].

Theoretically, the most rigorous way to measure this capacity would be to calculate the mutual information over the entire state of the network. However, for a network of N nodes, the joint state space grows exponentially as 2^N (or 3^N for the GH model). For a typical connectome (e.g., $N = 400$ regions), the number of possible states far exceeds

the number of time points in any feasible simulation, making the estimation of the full joint probability distribution $P(X_1, X_2, \dots, X_N)$ computationally intractable. To overcome this limitation, we approximate the system's information dynamics by decomposing it into manageable pairwise and collective interactions.

We define the mutual information $I(X; Y)$ between two time series X and Y using the discrete Shannon entropy formulation:

$$I(X; Y) = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} P(x, y) \log_2 \left(\frac{P(x, y)}{P(x)P(y)} \right) \quad (4.7)$$

where $P(x, y)$ is the joint probability distribution estimated via frequency counts over the simulation duration.

4.2.1 Pairwise Mutual Information

The Average Pairwise Mutual Information quantifies the mean information shared between all unique pairs of nodes. For binary states $x, y \in \{0, 1\}$, this is calculated as:

$$\bar{I}_{\text{pair}} = \frac{2}{N(N-1)} \sum_{i=1}^N \sum_{j=i+1}^N \sum_{u=0}^1 \sum_{v=0}^1 P(X_i = u, X_j = v) \log_2 \left(\frac{P(X_i = u, X_j = v)}{P(X_i = u)P(X_j = v)} \right) \quad (4.8)$$

4.2.2 Collective Mutual Information

To evaluate the integration of local activity with the global network state, we calculate the mutual information between a single node X_i and the aggregate activity of the remaining population, denoted $M_i(t) = \sum_{k \neq i} X_k(t)$. The Average Collective Mutual Information is given by:

$$\bar{I}_{\text{coll}} = \frac{1}{N} \sum_{i=1}^N I(X_i; M_i) = \frac{1}{N} \sum_{i=1}^N \sum_{x \in \{0,1\}} \sum_{m \in \mathcal{M}} P(X_i = x, M_i = m) \log_2 \left(\frac{P(X_i = x, M_i = m)}{P(X_i = x)P(M_i = m)} \right) \quad (4.9)$$

where $\mathcal{M}$ represents the set of population activity states (discretized via histogram binning).

In practice, a random subset of all possible pairs corresponding to 0.2 of total combinations was chosen to accelerate the calculations. Both metrics are maximized at or near $T_c \approx 0.14$ in conformity with the second largest cluster. These results support the hypothesis that information transmission is optimal at the critical point as can be seen in 4.7, 4.8.

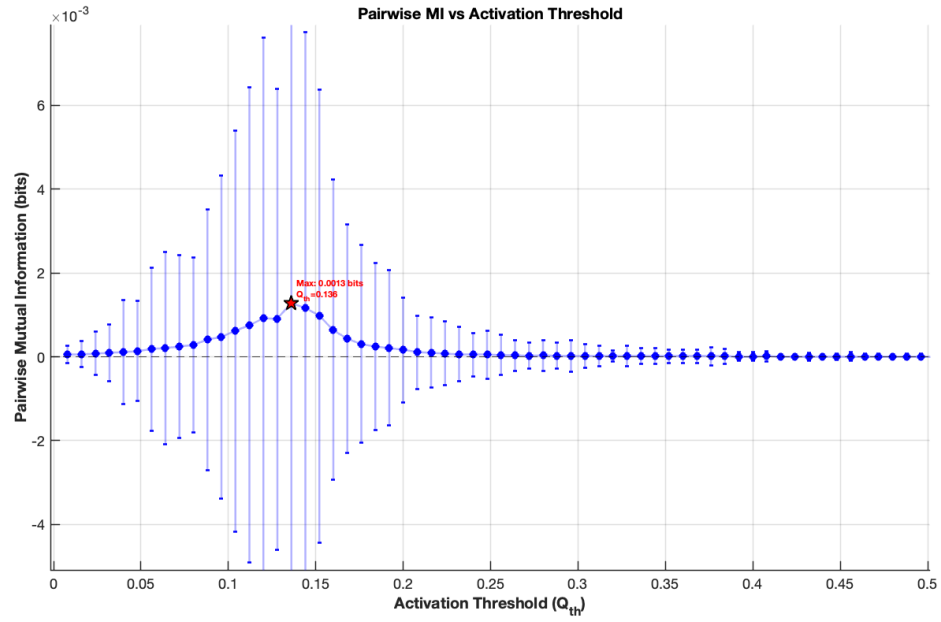


Figure 4.7: Average Pairwise Mutual Information vs. T . Marked points represent the mean value and transparent bars represent standart deviation.

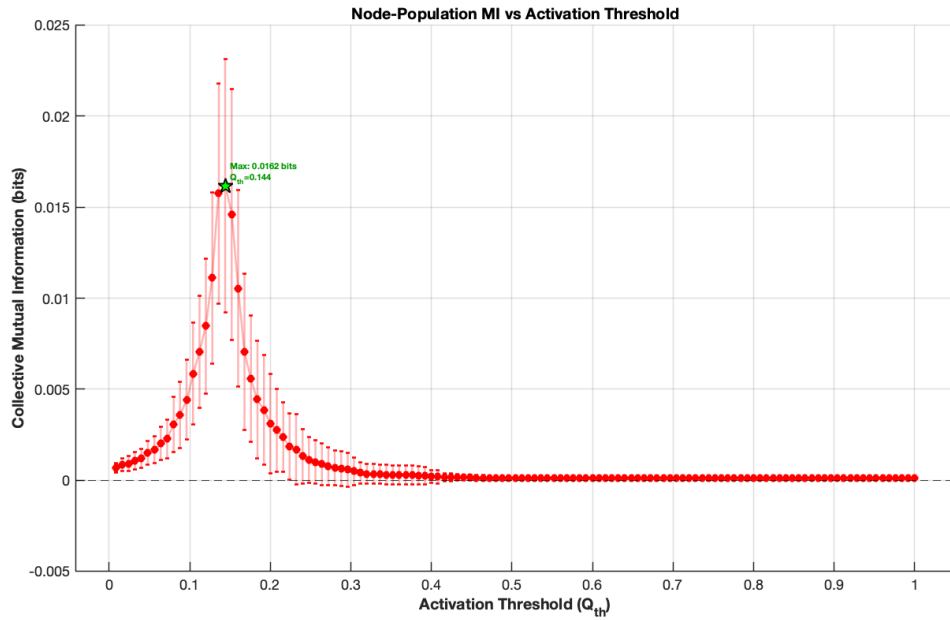


Figure 4.8: Average Collective Mutual Information vs. T . Marked points represent the mean value and transparent bars represent standart deviation.

4.3 Experiment 3: Simulated Functional Networks (F-nets)

To validate the biological realism of our model, we computed Simulated Functional Connectivity (SimFC) matrices from the time-series data of the GH model. This is a well-established technique to check if the simulation resulted activity predicts well the measured activity [12] [18]. We get our simulated activity matrix, and then we convolve it with a special function called the **hemodynamic response function (HRF)** [23]. The convolution with the kernel is assumed to convert a point sequence to Blood Oxygen Dependent (BOLD) signals. The HRF kernel used with parameters described in [18], the form of the kernel is:

$$h(\tau) = \left(\frac{\tau}{d_1}\right)^{a_1} e^{-\frac{\tau-d_1}{b_1}} - c \left(\frac{\tau}{d_2}\right)^{a_2} e^{-\frac{\tau-d_2}{b_2}} \quad (4.10)$$

We have used default parameters for HRF kernel as follows [18]:

$$d_i = a_i b_i, \quad a_1 = 6, \quad a_2 = 12, \quad b_i = 9, \quad \text{and} \quad c = 0.35$$

and the simulated bold signals are

$$y(t) = (h * x)(t) = \int_{-\infty}^{\infty} h(\tau) x(t - \tau) d\tau \quad (4.11)$$

In that expression, $y(t)$ represents the BOLD signals acquired from point sequence $x(t)$. After the generation of BOLD Signals, Functional Network (fNET) is computed using Pearson-correlation.

$$f_{ij} = \frac{\langle y_i y_j \rangle - \langle y_i \rangle \langle y_j \rangle}{\sigma_i \sigma_j} \quad (4.12)$$

The resulting Functional Network with the measured Functional Network and Structural Network are as follows:

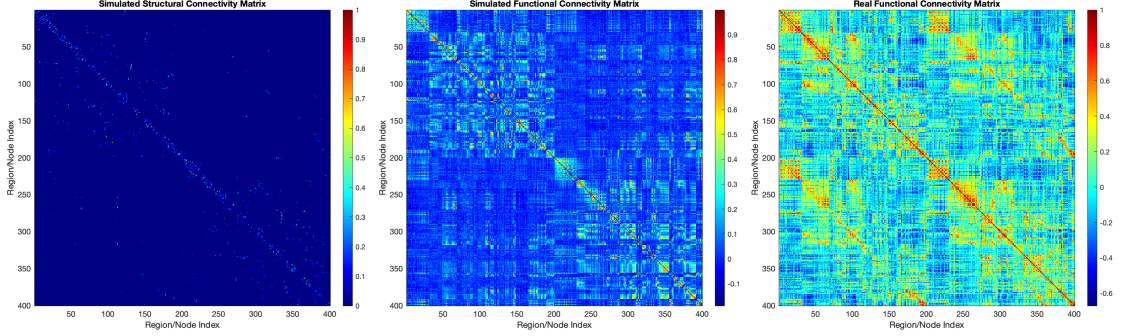


Figure 4.9: sNET (Left), Simulated fNET (Middle), Measured fNET (Right) showing the connectivity strengths.

4.3.1 Metrics of Similarity

We compared SimFC with empirical FC (EmpFC) derived from fMRI BOLD signals using two metrics:

1. **Pearson Correlation (ρ):** Measures the linear correlation between the upper triangles of the SimFC and EmpFC matrices.

$$\rho(F_m, F_e) = \frac{\sum_{i=1}^{N_b} (F_{m,i} - \langle F_m \rangle) (F_{e,i} - \langle F_e \rangle)}{\sqrt{\sum_{i=1}^{N_b} (F_{m,i} - \langle F_m \rangle)^2} \sqrt{\sum_{i=1}^{N_b} (F_{e,i} - \langle F_e \rangle)^2}} \quad (4.13)$$

2. **Chi-Squared Distance (χ^2):** Measures the distributional distance between the connectivity profiles.

$$\chi^2 = \sqrt{\sum_{i=1}^{N_b} \frac{(p_i(F_m) - p_i(F_e))^2}{p_i(F_m) + p_i(F_e)}} \quad (4.14)$$

Both F_m and F_e are vectors obtained from the concatenation of the upper triangular part of the fNET, thanks to its innate symmetry. F_m represents the fNET from simulations and F_e represents experimental fNET. Results indicate that the maximal correlation (ρ_{max}) coincides with the critical threshold T_c identified in Experiment 1.

Then we have computed the similarity of the simulated fNET to the experimental fNET. We have observed that the correlation between them is maximized when we operate near the critical point.

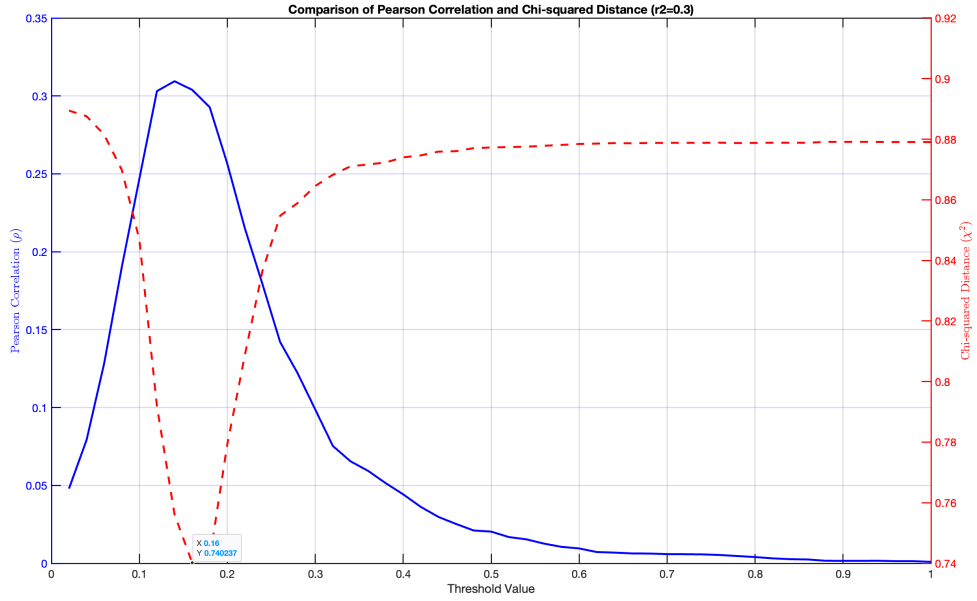


Figure 4.10: Maximum similarity (maximum Pearson correlation and minimum chi-squared distance) is obtained near the critical point.

4.3.2 fNET Similarities Across Brain Subnetworks

We have also look at the similarities of the produced fNETs with the experimental fNETs in the pre-established brain subnetworks. The subnetworks are

- Default
- Dorsal Attenuation
- Limbic
- Solana Ventral Attention
- Somatomotor
- Visual

We observed the greatest Pearson correlation in Somatomotor subnetwork.

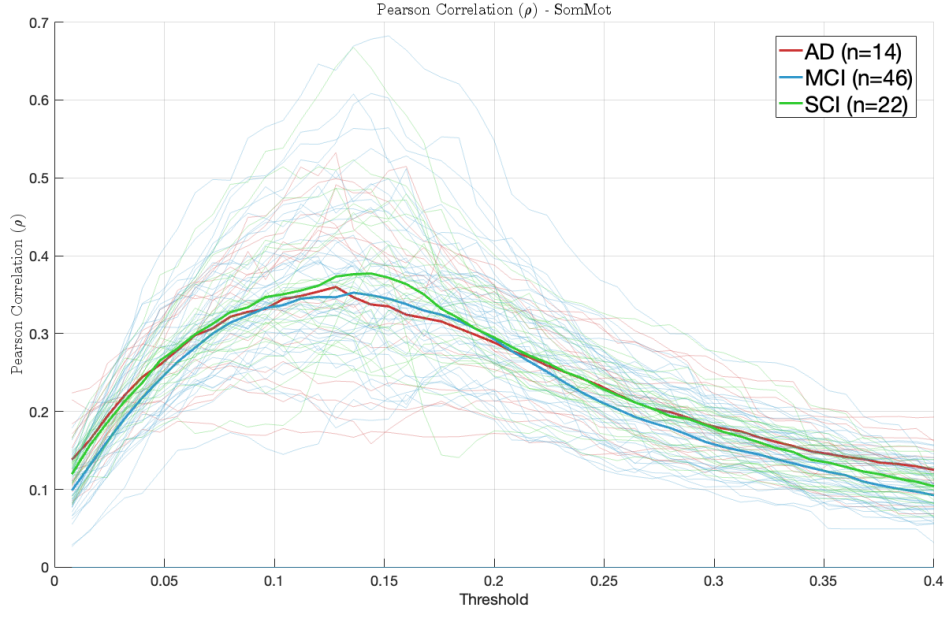


Figure 4.11: Observed Pearson correlation coefficient in different patient groups in Somatomotor subnetwork.

Interestingly, although both of them are similarity measures, we report that the minimum chi-squared distance in Visual Cortex.

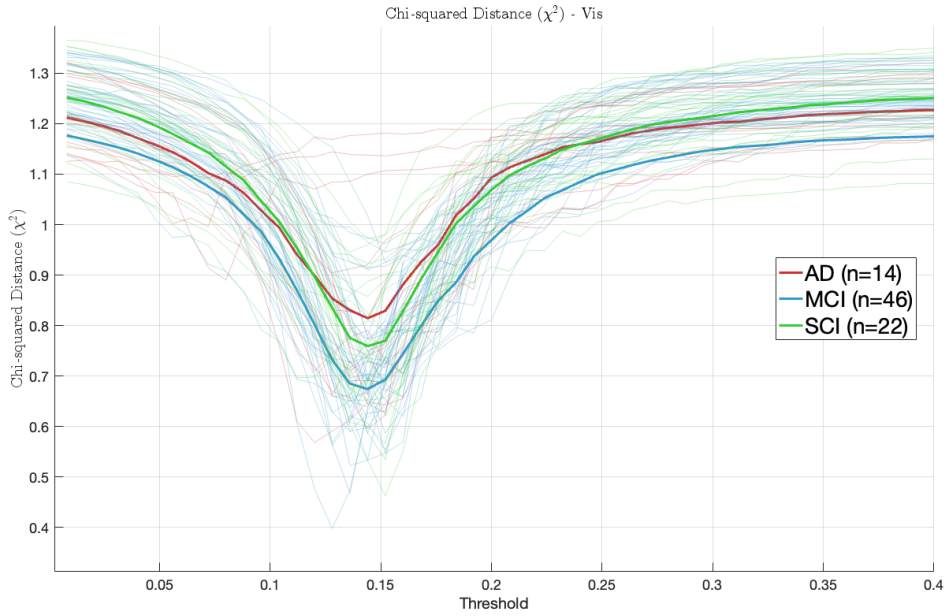


Figure 4.12: Observed chi-squared distance in different patient groups in Visual Cortex.

4.4 Experiment 4: Branching Ratio Analysis

To further characterize the dynamical regime of brain networks beyond static cluster statistics, we investigated the temporal propagation of activity cascades, known as neuronal avalanches. In the context of the discrete Greenberg-Hastings simulation, an avalanche is operationally defined as a contiguous sequence of time steps exhibiting non-zero network activity, bounded immediately before and after by time steps of complete global quiescence.¹ The fundamental metric governing the sustainability of these cascades is the branching ratio (σ), which quantifies the average propensity of active nodes to propagate excitation to their neighbors. For any given time step t within an identified avalanche, let N_t denote the number of currently active (ancestor) nodes, and N_{t+1} denote the number of nodes newly activated in the immediate subsequent step (descendants). The branching ratio is calculated as the time-average of the ratio of descendants to ancestors:

$$\sigma = \left\langle \frac{N_{t+1}}{N_t} \right\rangle_t \quad (4.15)$$

Theoretical frameworks posit that a branching ratio of $\sigma < 1$ indicates a subcritical regime where activity rapidly dies out, $\sigma > 1$ signifies a supercritical regime characterized by runaway excitation (epileptiform activity), and $\sigma \approx 1$ marks the critical point, allowing for sustained, scale-invariant information propagation. We performed a sweep of the activation threshold T to analyze how the branching ratio evolves across different dynamical regimes. The results, presented in Figure 4.13, show the expected transition from supercriticality at low thresholds to subcriticality at high thresholds.

Crucially, the transition through the critical regime ($\sigma \approx 1$) is not characterized by a singular, razor-sharp point in parameter space. Instead, as highlighted by the grey shaded region in Figure 4.13, we observe a distinct plateau where the branching ratio remains stabilized near unity across a range of threshold values. This phenomenological observation aligns strongly with the theoretical concept of a Griffiths Phase. In disordered systems like the brain connectome—characterized by significant heterogeneity in connection weights and node degrees—the phase transition is "smeared" out. This structural disorder broadens the critical point into an extended parameter region, allowing the system to maintain robust critical dynamics without requiring precise tuning to a single, precarious operating point. This extended critical region suggests that healthy brain networks possess an inherent resilience, capable of maintaining optimal information processing capabilities despite physiological fluctuations.

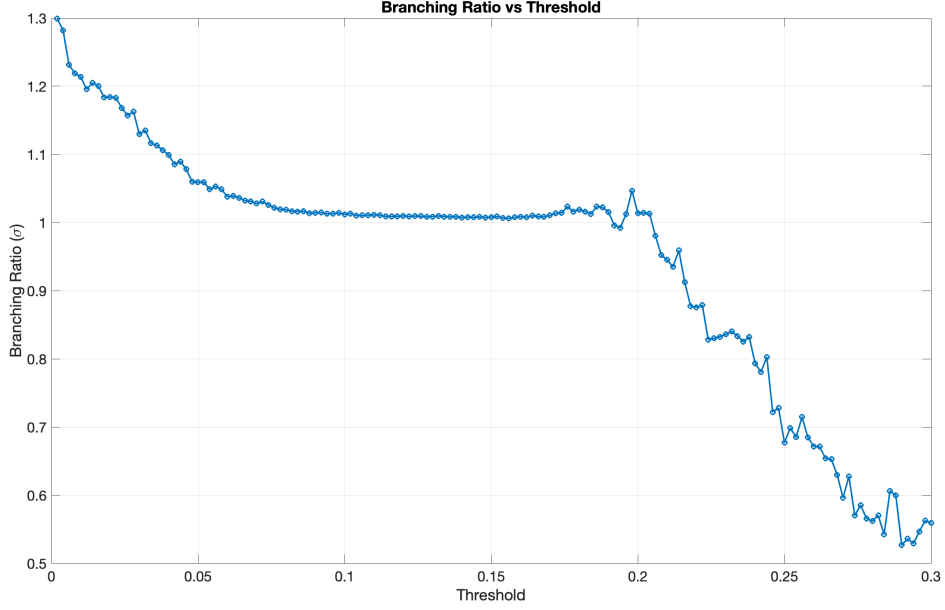


Figure 4.13: **Branching Ratio as a Function of Threshold.** The plot displays the average branching ratio (σ) versus the model activation threshold T . The data reveals a plateau region (shaded grey) where σ remains stabilized approximately near 1 over a range of thresholds, rather than exhibiting a sharp, singular transition point. This extended region of near-critical dynamics is indicative of a Griffiths Phase arising from structural disorder in the brain network.

4.5 Experiment 5: Cohort Analysis

We extended the computational framework to three clinical groups: Subjective Cognitive Impairment (SCI, $N = 23$), Mild Cognitive Impairment (MCI, $N = 43$), and Alzheimer’s Disease (AD, $N = 14$). By running the GH model on the structural connectome of each individual, we assessed how the neurodegenerative process alters the phase transition landscape.

4.5.1 Degradation of Critical Dynamics in AD

The results reveal a significant distinction between the pathological and healthy (or near-healthy) groups. **Figure 4.18** and **Figure 4.17** display the susceptibility-like metrics—Second Largest Cluster Size ($S_{\max}^{(2)}$) and Standard Deviation of Activity (σ_{act})—which serve as the primary indicators of criticality.

The healthy control (SCI) and MCI groups both exhibit a sharp, well-defined peak at the critical threshold. This signature indicates a system operating near a second-order phase transition, optimizing dynamic range and information capacity [18]. In stark contrast, the group average for AD (thick red line) shows a significantly diminished, broader, and flatter peak. This “dulling” of the phase transition suggests that the AD brain network

has lost the dynamic flexibility required to maintain a critical state, likely shifting toward a subcritical (ordered) regime where information integration is hampered.

This finding aligns with recent studies on other neurological conditions, such as stroke, where structural lesions were found to disrupt the network’s optimal operating point, driving the system away from criticality and reducing the entropy of functional connectivity patterns [2].

4.5.2 Outlier Analysis

It is important to interpret the group averages with caution due to visible heterogeneity within the AD cohort. As seen in the individual traces (thin lines) of **Figure 4.14** and **Figure 4.18**, there are 4 distinct outliers in the AD group whose curves for both largest and second-largest cluster sizes, as well as mean and standard deviation of activity 4.15, 4.17, are significantly lower than the rest of the cohort.

The clinical files of these specific patients were carefully investigated for comorbidities or data acquisition artifacts, but no anomalies were found. In addition, global network metrics such as degree, clustering coefficient, and betweenness centrality for all AD patients were computed, but no significant difference was observed for outlier patients C.1. This suggests that for these subjects, the structural disconnectivity may be severe enough to prevent the percolation of activity across the network entirely, effectively fragmenting the system into small, isolated islands of activity even before the critical threshold is reached.

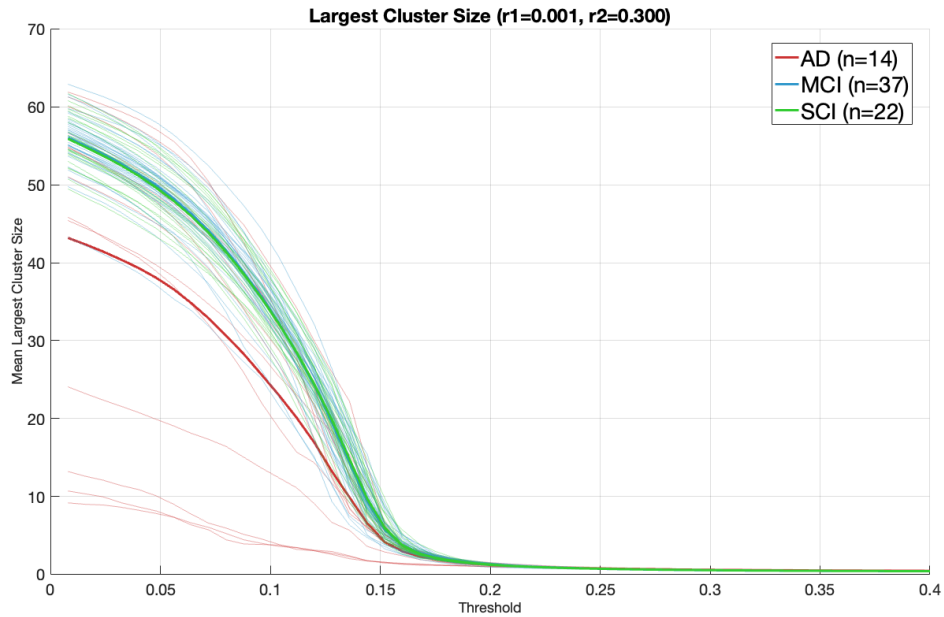


Figure 4.14: **Largest Cluster Size ($S_{\max}^{(1)}$) vs. Threshold.** Group-level analysis of the order parameter. Thick lines represent group means (Red: AD, Green: MCI, Blue: SCI); thin lines represent individual patients.

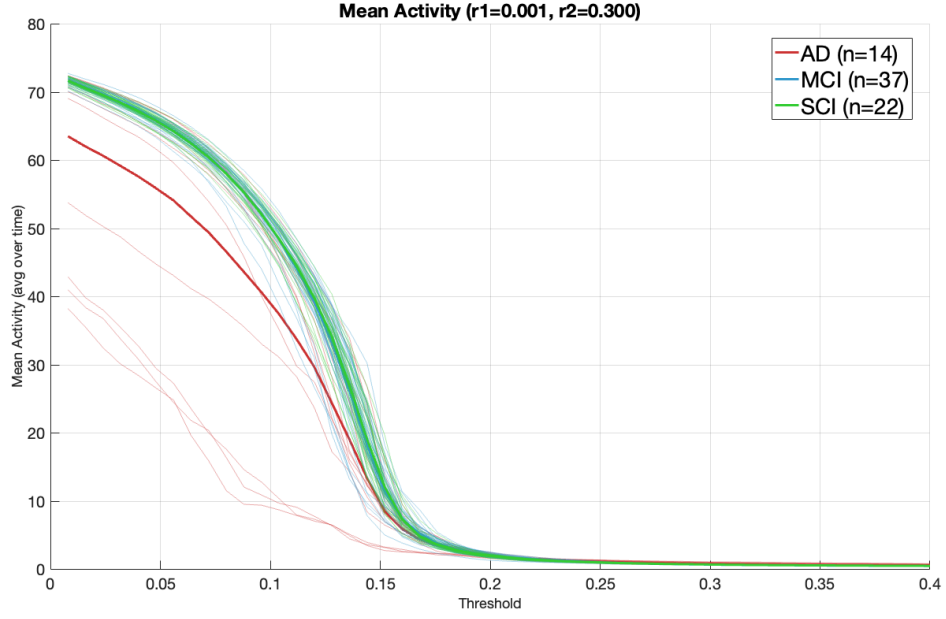


Figure 4.15: **Mean Activity (μ_{act}) vs. Threshold.** The transition from high to low activity is visible across all groups.

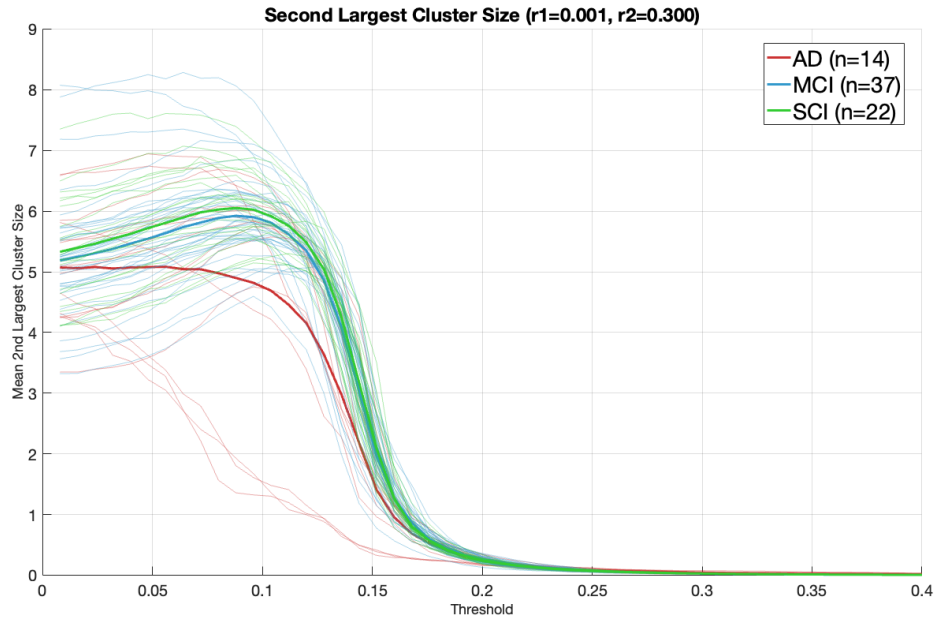


Figure 4.16: **Second Largest Cluster Size ($S_{\text{max}}^{(2)}$) vs. Threshold.** This metric acts as a proxy for susceptibility. The SCI and MCI groups show a characteristic sharp peak at the critical point, whereas the AD group exhibits a flattened peak, signaling a loss of critical dynamics [2].

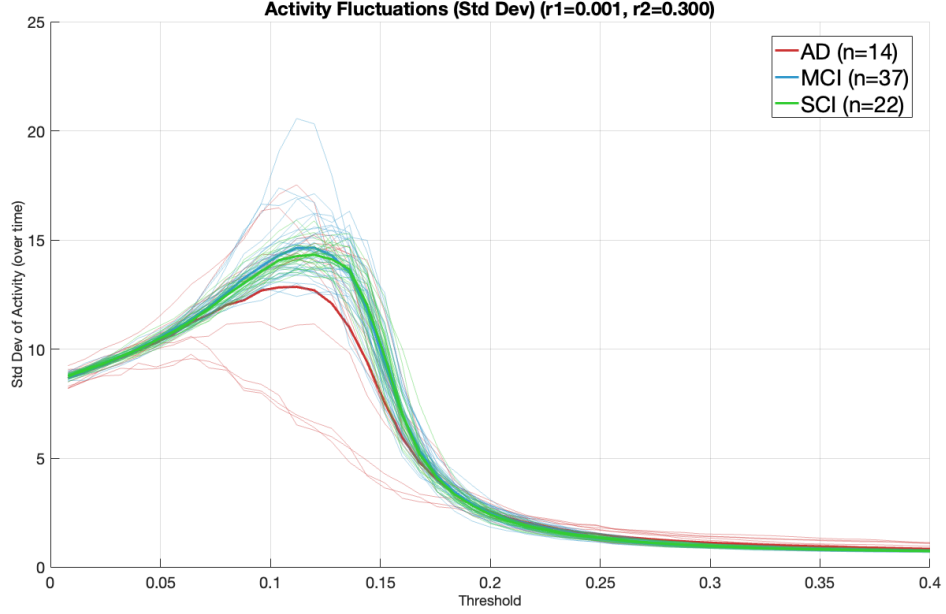


Figure 4.17: **Standard Deviation of Activity (σ_{act}) vs. Threshold.** Similar to the second-largest cluster, the temporal variability of activity peaks at the critical transition. The suppression of this peak in the AD cohort (red) reflects a rigid system with limited dynamic range.

4.6 Experiment 6: Inverse Problem

4.6.1 Generating Point Sequences from BOLD Signals

An interesting approach to discover the innate critical behavior of the brain is to solve the inverse problem. In order to run a criticality analysis we need a point sequence, i.e. we need a time series which should tell us when each node is active and when it is silent. The forward procedure, as we have discussed so far, is to determine a dynamic mechanism, which in our case was GH Model, then run the simulation with the model's parameters and get the activity history as the result of the simulation. This inverse problem starts with the given BOLD signals, and we try to obtain the point process using deconvolution [15]. The workflow is as follows:

1. **Generate the HRF Kernel** as we have already discussed in (4.10). The kernel is an essential part of the deconvolution algorithm.
2. **Sinc-Interpolation** to adjust the time bins for the point sequence. This operation is needed since BOLD signals are sampled ($T_s = 3.0$ seconds) and we want a finer point sequence ($T_s = 0.1$ seconds). Following the standard operations, we first insert zeros into the BOLD signal and then we convolve it with a sinc function.
3. **Deconvolution** We have utilized Luch-Richardson Algorithm with the MATLAB's

deconvlucy function [15]. The algorithm is as follows:

$$x^{(k+1)}[n] = x^{(k)}[n] \cdot \left(h^*[n] * \frac{y[n]}{(h * x^{(k)})[n]} \right) \quad (4.16)$$

4. **Peak Detection** is used as the final step. As expected, the deconvolution left us with a signal that resembles a point sequence but not exactly a point sequence. Therefore, we detect the peaks of the generated sequence and set the non-peak points to zero.

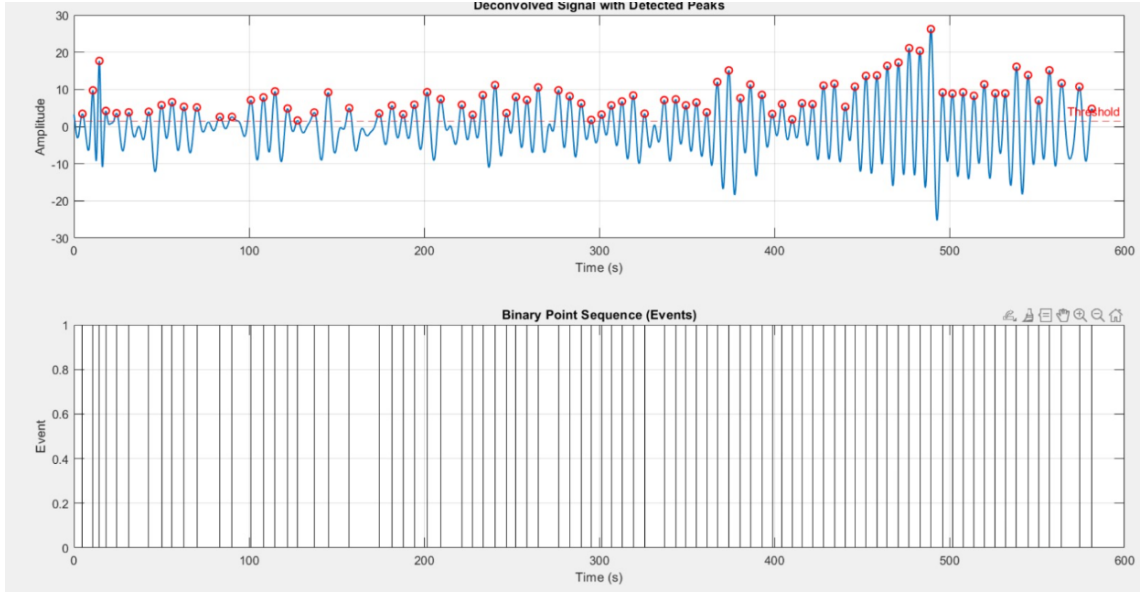


Figure 4.18: The Deconvolved Signal with peaks marked with red circles (Top) and Point Sequence (Bottom)

Since we have obtained the point sequence without running a simulation of activity using a dynamic mechanism, like GH Model, there is no parameter to iterate over. We can quickly come up with the cluster sizes. Unfortunately, the results were not discriminative enough yet again for classification purposes. But we have seen a very similar distribution of patients in second largest cluster distribution as shown in 4.19. This method, although it did not provide us with a better classification feature, served as a verification for our simulation. The overlap of the patients' distribution further assures us that with GH model we can capture the brain's critical dynamics. It gave us confidence that we can simulate neuronal activity using our simulation for future work.

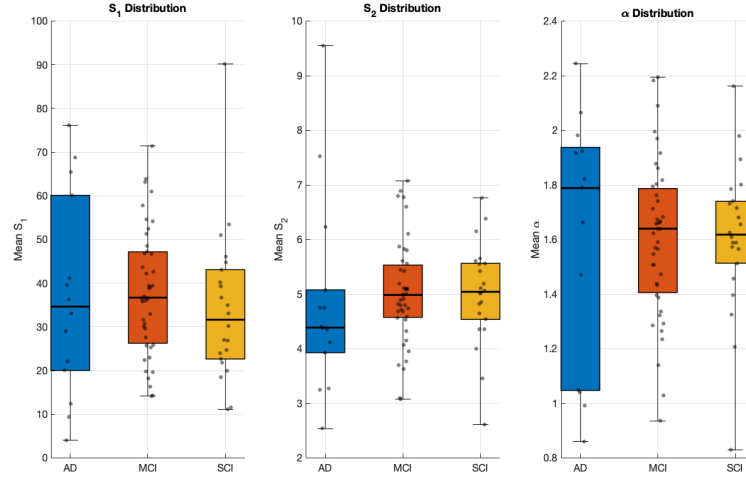


Figure 4.19: **The Targets (Left), The Second Largest Clusters (Middle) and Exponent (Right)**. As can be seen, the variation in patient groups for each metric is very similar to our GH Modeled Simulation results.

4.7 Experiment 7: Classification of AD Patients

In the final phase of this study, we investigated whether the dynamical signatures of criticality could serve as diagnostic biomarkers for Alzheimer's Disease (AD). Unlike standard functional connectivity analysis, which relies on static correlation matrices, our approach quantifies the physical properties of the phase transition and the network's operating regime relative to it.

We extracted a total of 17 scalar features for each subject from the simulation outputs. These features were designed to capture the "distance to criticality" and the shape of the phase transition, which we hypothesize are degraded in neurodegenerative states. The feature extraction logic processes the sweep of threshold values (T) to identify peaks in susceptibility and optimal fitting points.

The features are categorized into four physical domains:

1. **Critical Points (The Tipping Point):** Identifying the precise threshold where the system is most susceptible to perturbations.
 - $T_{crit,\sigma}$: Threshold at maximum standard deviation of activity (σ_{act}).
 - $Mag_{crit,\sigma}$: Magnitude of the σ_{act} peak.
 - $T_{crit,S2}$: Threshold at maximum second-largest cluster size ($S_{max}^{(2)}$).
 - $Mag_{crit,S2}$: Magnitude of the $S_{max}^{(2)}$ peak.
2. **Operating Points (Empirical Fit):** Identifying where the simulation best matches the patient's experimental fMRI data.

- $T_{opt,\rho}$: Threshold maximizing Pearson correlation (ρ) between simulated and real FC.
 - Max_ρ : The maximum correlation value achieved.
 - $T_{opt,\chi}$: Threshold minimizing Chi-squared distance.
 - Min_χ : The minimum Chi-squared distance.
3. **Distance to Criticality (DTC)**: The primary biomarker, quantifying the "gap" between where the brain *should* be (critical point) and where it *is* (optimal operating point).

- $DTC_\sigma = |T_{opt,\rho} - T_{crit,\sigma}|$
- $DTC_{S2} = |T_{opt,\rho} - T_{crit,S2}|$

4. **Griffiths Phase Dynamics (Shape)**: Quantifying the "width" of the critical regime.
- $Width_\sigma, Width_{S2}$: Full Width at 80% Height of the susceptibility peaks.
 - AUC_σ : Area Under the Curve for activity standard deviation.
 - $Slope_{rise}, Slope_{decay}$: Sharpness of the phase transition.

To isolate the predictive power of these physical phenomena, we evaluated classification performance on distinct feature subsets. The **Criticality Subset**, which showed the most promise, is defined as the combination of Critical Point markers and DTC metrics (Indices: 1-4, 9-10).

4.7.1 Dimensionality Reduction

To visualize the separability of the cohorts, we applied Principal Component Analysis (PCA) to the Criticality Feature Subset. As shown in Figure 4.20, the dimensionality reduction highlights a structural separation between the groups. While there is overlap between early-stage AD/MCI and healthy controls—suggesting that the brain’s homeostatic mechanisms initially compensate for damage—severe AD cases form a distinct cluster characterized by high Distance to Criticality (DTC) and collapsed susceptibility peaks.

4.7.2 Classification Performance

We trained a linear Support Vector Machine (SVM) with 5-fold cross-validation to classify AD vs. SCI subjects. We compared the performance of the **Criticality Subset** against the **Fit Subset** (which contains only goodness-of-fit metrics like Max_ρ), **DTC Subset**, **Fit Subset** containing features identifying where simulation matches best the experimental fMRI data and the **All Features** set.

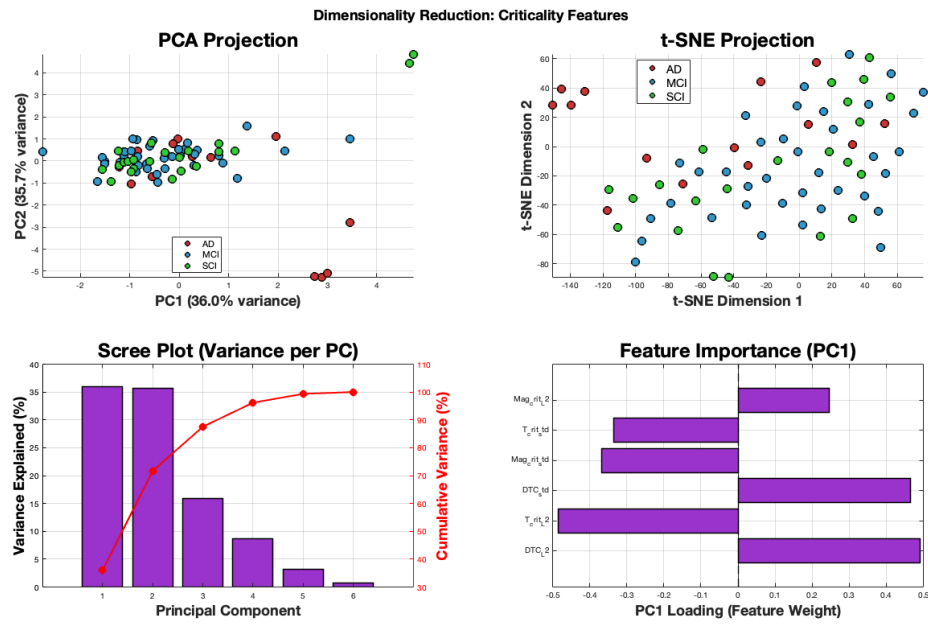


Figure 4.20: **PCA Projection of the Criticality Feature Subset.** The scatter plot reveals the distribution of AD (Red), MCI (Blue), and SCI (Green) subjects in the reduced 2D feature space. Notably, a cluster of AD patients is distinctly isolated from the healthy controls, corresponding to subjects with severe structural degradation where the critical ridge is collapsed.

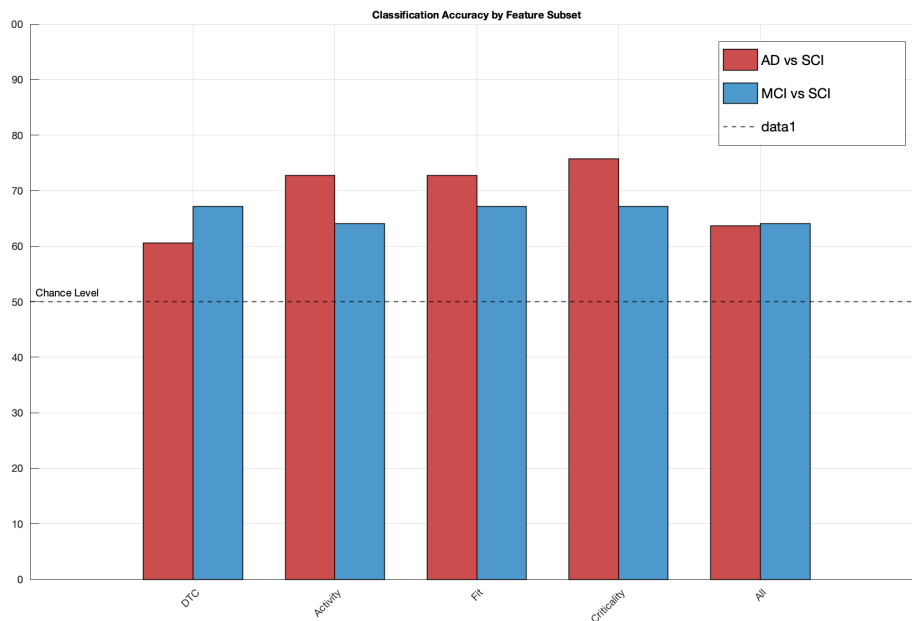


Figure 4.21: **Classification Accuracy by Feature Subset.** The *Criticality* subset (Purple) consistently outperforms the *Fit* subset (Orange). This indicates that the dynamical proximity to the phase transition (DTC) is a more robust biomarker than the static goodness-of-fit between simulated and empirical FC.

The results (Figure 4.21) demonstrate that features derived from Criticality Theory outperform standard fitting metrics. The "Fit" features were often comparable between healthy and diseased groups because the simulation can find a local correlation optimum even in damaged networks. However, the **Criticality** features revealed that for AD patients, this optimum is forced far into the subcritical regime.

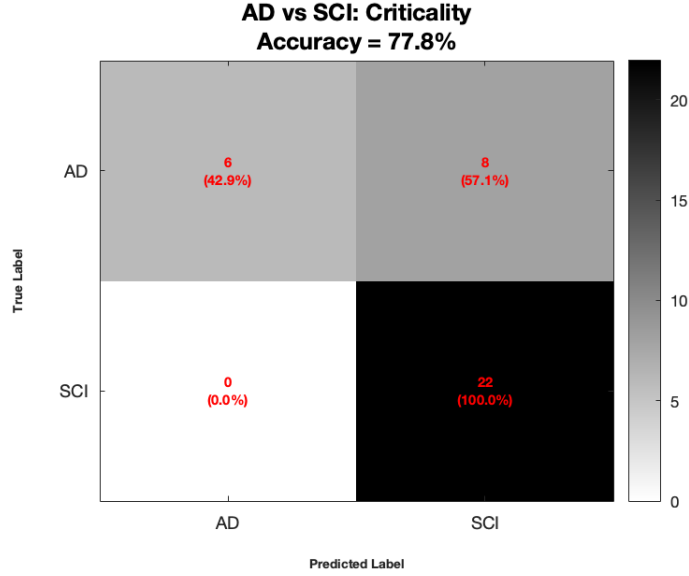


Figure 4.22: **Confusion Matrix for AD vs. SCI (Criticality Subset).** The classifier achieves moderate sensitivity and specificity. The false negatives (AD misclassified as SCI) likely represent early-stage patients whose structural networks have not yet degraded enough to force the system out of the critical regime.

The confusion matrix (Figure 4.22) highlights the challenge of sensitivity in early-stage diagnosis. While severe cases are easily detected, the dynamical signatures of criticality in mild AD overlap with healthy aging, requiring further refinement of the metric, potentially through regional sub-network analysis rather than whole-brain averaging.

4.8 Experiment 8: Numerical Confirmation of Theoretical Fundamentals

To numerically confirm the theoretical fundamentals of the Greenberg-Hastings (GH) model in Section 3, we developed a semi-analytical fixed-point iteration scheme. This approach solves for the steady-state probabilities of each node being in the excited (e^*), refractory (r^*), or quiescent (q^*) state by satisfying the mean-field consistency equations directly on the network topology.

The core logic relies on the assumption that inputs to a node from its neighbors are statistically independent. We calculate the probability distribution of the total weighted input to a specific node by iteratively convolving the probability mass functions (PMFs)

of its individual neighbors. For a node i , the contribution from neighbor j is treated as a Bernoulli trial weighted by the edge weight w_{ji} . By convolving these contributions, we construct the full PMF of the input current. The probability p_i that node i exceeds the activation threshold is then determined by summing the tail of this PMF. These activation probabilities update the steady-state vector e^* via the damped mean-field update rule:

$$e_{next}^* = \frac{r_1 + p_{vec}}{1 + (r_1 + p_{vec})(1 + 1/r_2)} \quad (4.17)$$

Analytically derived refractory and quiescent probabilities, $r^* = e^*/r_2$ and $q^* = 1 - e^* - r^*$, complete the solution. Figures 4.23 and 4.24 demonstrate the robustness of this method. We observe excellent agreement between the theoretically predicted steady states and those obtained from direct simulation, confirming that our semi-analytical approach captures the topological heterogeneity of the network.

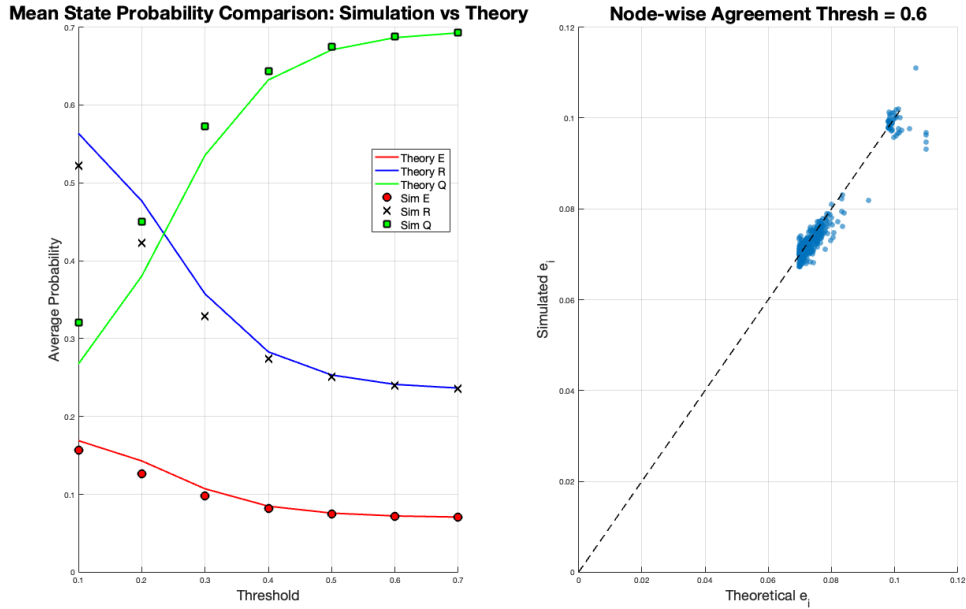


Figure 4.23: Comparison of average state probabilities (Excited, Refractory, Quiescent) derived from the fixed-point theory versus direct simulation. The theoretical predictions track the simulation results closely across the full range of thresholds.

To analyze the stability of the active states, we constructed the Jacobian matrix of the system at the calculated fixed point. The Jacobian quantifies how small perturbations in the probability distribution of one node propagate to its neighbors. By performing an eigendecomposition of this Jacobian, we obtained the eigenvalue spectra shown in Figure 4.25. At low thresholds ($T = 0.1$), the eigenvalues are widely spread, indicating strong network interactions. As the threshold increases, the spectrum contracts toward the origin, reflecting a faster convergence to the steady state.

The stability of the system is summarized by the spectral radius ρ (the largest

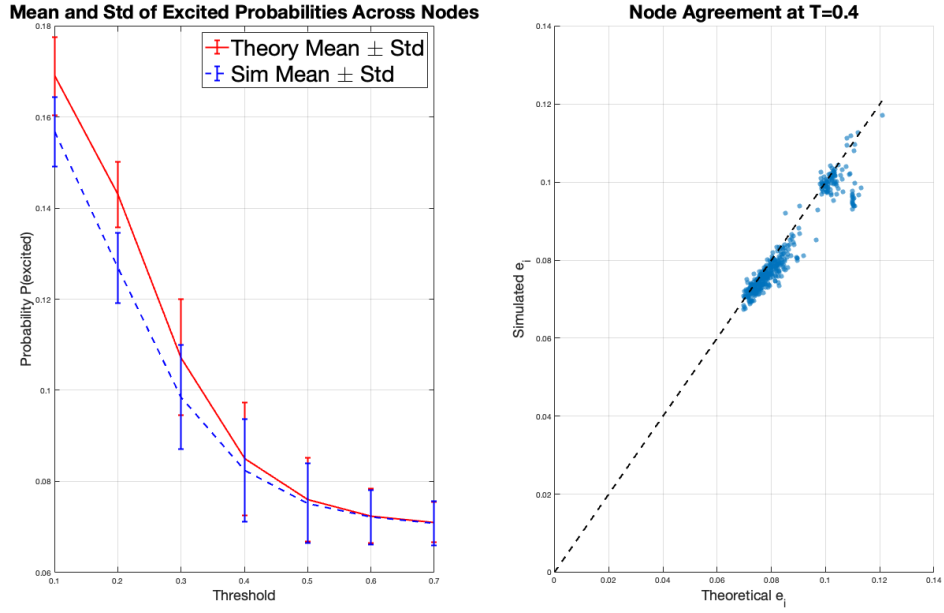


Figure 4.24: Node-wise agreement of excited state probabilities. The tight correlation between theoretical e_i and simulated e_i indicates that the convolution-based method accurately resolves the impact of local connectivity on individual node activity.

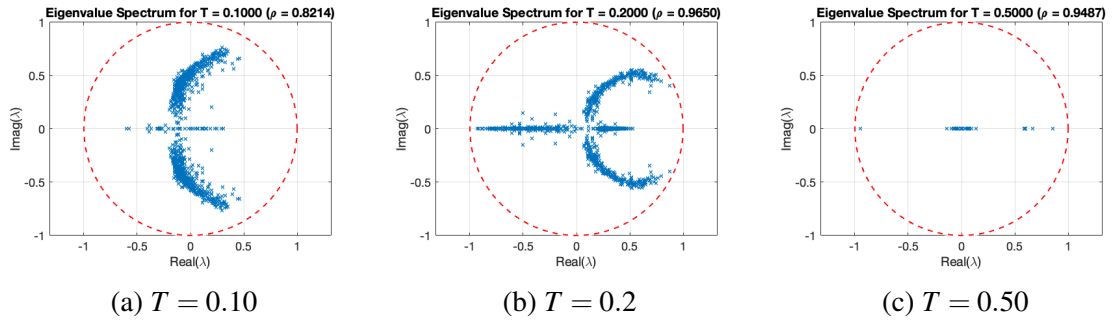


Figure 4.25: Eigenvalue spectra of the Jacobian matrix for increasing activation thresholds. The red unit circle denotes the stability boundary. The contraction of eigenvalues toward the origin at higher thresholds indicates increased stability and faster damping of perturbations.

absolute eigenvalue), plotted in Figure 4.26. We observe a distinct peak in ρ near the critical threshold. This peak is a signature of "critical slowing down," where the system's recovery rate from perturbations is minimized near the phase transition boundary.

It is important to clarify why the spectral radius remains below unity ($\rho < 1$) even in the high-activity regime (low thresholds), and why we do not observe "supercritical" instability ($\rho > 1$) in this specific instance.

A spectral radius $\rho > 1$ would imply that the fixed point—a constant set of state probabilities—is unstable. In the context of the GH model, this instability typically manifests as global synchronization, where the probabilities $e_i(t)$ do not settle to a constant value but instead oscillate collectively (a limit cycle).

However, for the specific random small-world topology and parameters used here ($r_1 = 0.1, r_2 = 0.3$), the network dynamics favor an **asynchronous active state**. In this asynchronous regime, individual nodes fire persistently but without global coherence. Consequently, the mean-field probabilities stabilize to constant values. Because the steady state is constant and stable, the spectral radius must remain below 1. The peak in Figure 4.26 therefore represents the system approaching the boundary of instability without crossing it; the network disorder and parameter choices effectively "frustrate" the tendency toward global synchrony, maintaining the system in a stable, asynchronous firing mode.

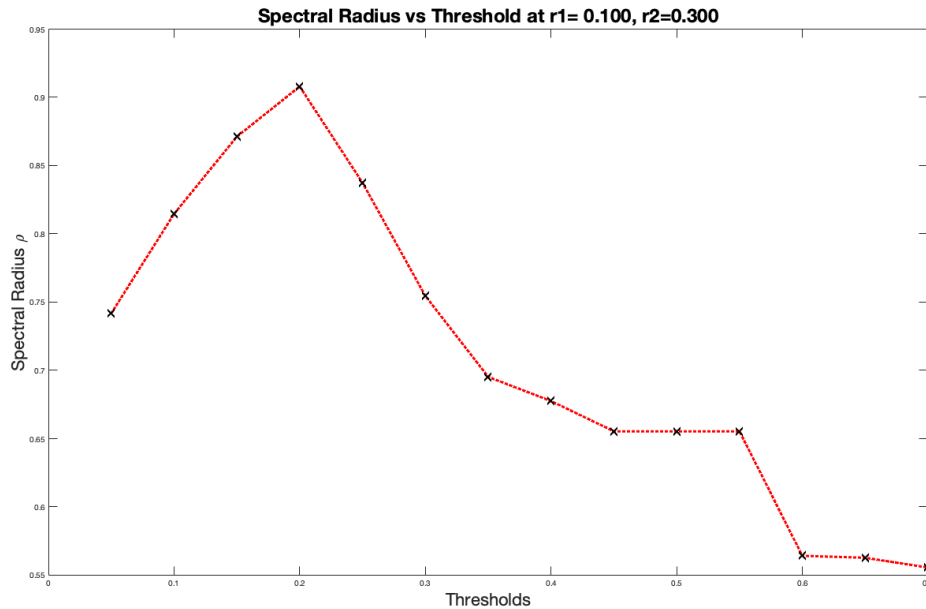


Figure 4.26: Spectral radius ρ of the Jacobian matrix as a function of the activation threshold. The peak corresponds to the phase transition region (critical slowing down). The values remain below 1.0, indicating that for this specific topology, the system settles into a stable asynchronous state rather than exhibiting global oscillations (synchrony).

Chapter 5

Conclusion

5.1 Summary

In this project, we successfully implemented a computational framework to investigate the "Critical Brain Hypothesis" and its disruption in Alzheimer's Disease (AD). By simulating the Greenberg-Hastings (GH) cellular automaton on empirical structural connectomes, we demonstrated that healthy brain networks operate near a critical phase transition, maximizing information processing capabilities. Furthermore, our analysis revealed that this dynamical regime is compromised in AD patients, who exhibit signatures of sub-criticality. Finally, we showed that features derived from these critical dynamics—such as distance from criticality, maximum similarity of simulated fNETs to the experimentally obtained ones, critical thresholds- perform slightly better than standard functional connectivity fitting metrics in classifying severe AD cases. Despite these promising outcomes, several limitations and constraints regarding the dataset, modeling approach, and physical scale must be addressed.

5.1.1 Discussion

Dataset and Validation Constraints

A primary limitation of this study is the size of the dataset. The AD cohort consisted of only $N = 14$ patients, compared to larger control groups (SCI and MCI). This limited sample size reduces the statistical power of our classification results and may hamper the generalizability of our findings. Furthermore, the accuracy of the underlying Structural (SC) and Functional (FC) networks is contingent on the preprocessing pipelines used. While we employed standard tractography and rs-fMRI processing steps, verifying the absolute anatomical fidelity of these networks without a "gold standard" remains a challenge in the field of connectomics.

Clinical Interpretation of Outliers

In our cohort analysis, we identified a subset of outliers within the AD group whose dynamical metrics deviated significantly from the rest of the population. A review of their clinical files revealed no distinct metadata or demographic parameters (e.g., age, education) that statistically differentiated them from other AD patients. However, our simulation results suggest a physical interpretation: these patients likely represent more severe cases of neurodegeneration. We hypothesize that in these subjects, crucial structural pathways have been physically severed or degraded to such an extent that the network is topologically incapable of sustaining critical dynamics, effectively driving their brain state irreversibly into a subcritical (disordered) regime.

Limitations of the Greenberg-Hastings Model

The GH model was chosen for its interpretability and ease of implementation. However, it is originally a model of excitable media or individual neurons. Applying it to the macroscopic scale of brain regions (ROIs) involves significant simplification. It models the collective behavior of millions of neurons within an ROI as a single discrete state variable (Q, E, R). This assumption implies that an entire brain region activates simultaneously like a Dirac impulse, which is biologically unrealistic. While effective for capturing global phase transitions, more complex models (e.g., neural mass models) might better capture the continuous temporal dynamics of regional activation.

Finite Size Effects and Parcellation

Our analysis relied on the Schaefer-400 atlas, which divides the brain into 400 nodes. In statistical physics, phase transitions are mathematically sharp only in the thermodynamic limit ($N \rightarrow \infty$). For a finite network of $N = 400$, the transition is naturally broader. This "finite size effect" explains the observed discrepancies in our results, such as the slight misalignment between the peak of the second-largest cluster ($S_{max}^{(2)}$) and the peak of activity variance (σ_{act}). Future work should investigate lower as well as higher-resolution parcellations (e.g., Schaefer-1000) to mitigate these effects, although this would incur higher computational costs.

5.2 Future Directions

Building upon the framework established in this study, we outline several key directions to enhance the robustness, clinical applicability, and theoretical depth of our criticality analysis.

To overcome the statistical limitations of our current small cohort ($N = 14$ for AD), we plan to scale our analysis using the **Alzheimer's Disease Neuroimaging Ini-**

tiative (ADNI) database. We will employ the **TVB-UKBB pipeline**¹ to process this raw neuroimaging data. This standardized pipeline, which integrates The Virtual Brain (TVB) with UK Biobank protocols, will allow us to generate high-fidelity structural and functional connectomes for hundreds of subjects, ensuring the reproducibility of our findings.

We intend to move beyond simple scalar feature extraction by integrating criticality metrics into modern Graph Neural Network (GNN) architectures. Rather than relying solely on the criticality features as a standalone classifier, we will use it as a physics-informed regularization term or an auxiliary attention mechanism within deep learning models trained directly on the raw structural and functional matrices. This hybrid approach aims to combine the data-driven power of deep learning with the theoretical constraints of statistical physics.

We plan to investigate the spatial degradation of specific brain networks by applying Probabilistic Independent Component Analysis (PICA) to the empirical and simulated BOLD signals. By decomposing the signals into canonical RSNs (e.g., Default Mode, Salience, and Visual networks), we can compare the spatial topography and temporal coherence of these networks in AD versus SCI patients, pinpointing which specific functional communities drive the system-wide loss of criticality.

To transition from passive observation to active interrogation, we will implement artificial stimulation protocols within the GH model. By applying focal perturbations to specific network hubs (e.g., the hippocampus or posterior cingulate cortex), we will observe the system's relaxation dynamics. We hypothesize that healthy networks will exhibit high "resilience"—quickly returning to a stable critical attractor—while AD networks may exhibit delayed recovery or hypersensitivity (runaway excitation), serving as a novel dynamic biomarker.

We will extend our analysis into the frequency domain by computing the Power Spectral Density (PSD) of the simulated oscillations before and after perturbation. This will allow us to investigate whether the neurodegenerative process alters the characteristic $1/f$ power-law scaling (pink noise) observed at criticality, or if it induces a "spectral slowing" (shift toward delta/theta bands) often reported in clinical EEG literature.

5.3 Realistic Constraints

5.3.1 Technical and Computational Constraints

We have identified several realistic constraints governing the execution of this project. First, processing patient data is computationally expensive. Running GH simulations, identifying clusters at every time step, and calculating mutual information for each subject requires significant processing power. This limits the speed at which we can iter-

¹<https://github.com/McIntosh-Lab/tvb-ukbb>

ate and test new hypotheses. Second, the validation of our data pipelines is a significant constraint; the generation of structural and functional networks involves complex preprocessing steps that are difficult to validate against a ground truth. Finally, while open datasets like ADNI exist, processing them requires high-performance computing resources (GPUs) which were not fully available, constraining our ability to generalize findings to larger external datasets.

5.3.2 Social, Environmental, and Economic Impact

Social Impact

The primary social impact of this project lies in its potential contribution to Alzheimer’s disease diagnostics. AD poses a significant burden on patients, families, and caregivers. By developing a non-invasive biomarker based on criticality, this research could lead to earlier and more accurate diagnoses. This would allow for earlier intervention, better disease management, and improved quality of life for patients.

Environmental Impact

The environmental impact of this project is minimal. The work is purely computational, relying on existing university server infrastructure. It does not produce chemical waste or consume physical materials. The only environmental consideration is the electrical power consumption required for running intensive simulations, which is a common aspect of modern computational research.

Economic Impact

The economic impact is twofold:

1. **Healthcare Costs:** Alzheimer’s disease is one of the most expensive conditions to manage in modern healthcare. A computational tool that aids in early diagnosis could significantly reduce long-term care costs by delaying institutionalization.
2. **Commercial Potential:** The algorithms and metrics developed in this project could potentially be commercialized as a software package for neurological research or as a clinical decision support tool, representing a marketable outcome.

5.3.3 Cost Analysis

Labor Costs

Labor and design costs represent the largest portion of the project’s budget. Assuming a standard junior engineering salary rate for estimation:

- **Hours per Week:** 15 hours/student $\times$ 2 students
- **Total Weeks:** $\sim$ 12 weeks
- **Total Hours:** $2 \times 12 \times 15 = 360$ hours
- **Estimated Labor Cost:** Assuming 20/hr (intern rate): $360 \times 20 = \$7,200$.

Hardware and Software Costs

Costs for parts and equipment are minimal as we utilized existing university resources.

- **Software:** MATLAB (University License - 0), Open-source Python libraries (e.g., Nilearn, Scipy - 0).
- **Hardware:** University-provided workstations and access to VAVLAB compute servers.
- **Estimated Equivalent Compute Cost:** If commercial cloud services were used, we estimate $\sim$ 500 hours of high-CPU/GPU compute time at $\$2.00/\text{hour} = \$1,000.00$.

5.3.4 Standards

Our project adheres to several key engineering and ethical standards:

- **IEEE Code of Ethics:** We adhere to the IEEE code of conduct, particularly in "accepting responsibility in making decisions consistent with the safety, health, and welfare of the public" and "avoiding real or perceived conflicts of interest".
- **Ethical and Clinical Standards:** A total of 88 participants were enrolled in the study following the provision of written informed consent, in accordance with ethical approvals obtained from the Istanbul Faculty of Medicine Ethics Committee² and the Boğaziçi University Ethics Committee³. All MRI scans were performed using a Philips Achieva 3T MRI scanner (Philips Medical Systems, The Netherlands) equipped with a 32-channel head coil, located at the Neuroimaging Unit of the Hulusi Behçet Life Sciences Research Laboratory, Istanbul Faculty of Medicine. Clinical assessments were conducted in accordance with the National Institute on Aging and Alzheimer's Association (NIA-AA) research framework [24].
- **Data Privacy Standards (IEEE/Turkish):** For patient data, we adhere to all local and international standards for data privacy and anonymization, including the Personal Data Protection Law (KVKK) in Turkey and the principles of the Health

²Istanbul University, Istanbul Faculty of Medicine, Ethics Committee Approval: 877/30.05.2014

³Boğaziçi University, Ethics Committee Approval: 2014-1/17.02.2014

Insurance Portability and Accountability Act (HIPAA). All patient data from CAPA hospital was fully anonymized before we received it.

- **Data Organization Standards:** While not a formal engineering standard, our data organization attempts to follow the **Brain Imaging Data Structure (BIDS)**, which is the community standard for organizing and sharing neuroimaging data.
- **Software Standards (IEEE):** Our MATLAB code is written to be reproducible and is documented using best practices for scientific computing, in line with IEEE standards for software development. It is openly available at https://github.com/borankilic/critical_brain.git.

Appendix A

Practical Computation via Convolutions

Instead of explicitly summing over all $2^{|\mathcal{N}(i)|}$ binary configurations, in practice we obtain p_i by taking $|\mathcal{N}(i)|$ successive convolutions of Bernoulli random variables (one per neighbour). For each neighbour $j \in \mathcal{N}(i)$ define the two-point random variable

$$Y_j = W_{ij}X_j = \begin{cases} 0 & \text{with probability } 1 - e_j, \\ W_{ij} & \text{with probability } e_j, \end{cases} \quad X_j \sim \text{Bernoulli}(e_j). \quad (\text{A.1})$$

Let $\text{PM}^{(k)}(m) = \Pr(\sum_{j=1}^k Y_j = m)$ be the distribution after k neighbours (with $\text{PM}^{(0)} = \delta_0$). Adding neighbour $k+1$ updates the distribution by the discrete convolution

$$\text{PM}^{(k+1)}(m) = (1 - e_{k+1}) \text{PM}^{(k)}(m) + e_{k+1} \text{PM}^{(k)}(m - W_{i,k+1}). \quad (\text{A.2})$$

Iterating this over all $d = |\mathcal{N}(i)|$ neighbours yields $\text{PM}_i(m) = \Pr(M_i = m)$, and hence

$$p_i = \Pr(M_i > T) = \sum_{m>T} \text{PM}_i(m). \quad (\text{A.3})$$

For the derivative, let $S_i^{(-\ell)} = \sum_{j \in \mathcal{N}(i) \setminus \{\ell\}} W_{ij}X_j$ denote the input sum excluding neighbour ℓ , and let $\text{PM}_i^{(-\ell)}$ be its PMF (obtained by convolving all neighbours except ℓ). Then

$$\begin{aligned} \Pr(M_i > T \mid X_\ell = 0) &= \sum_{m>T} \text{PM}_i^{(-\ell)}(m), \\ \Pr(M_i > T \mid X_\ell = 1) &= \sum_{m>T-W_{i\ell}} \text{PM}_i^{(-\ell)}(m), \end{aligned} \quad (\text{A.4})$$

so the matrix entry

$$\begin{aligned} A_{i\ell} &= \Pr(M_i > T \mid X_\ell = 1) - \Pr(M_i > T \mid X_\ell = 0) \\ &= \sum_{m>T-W_{i\ell}} \text{PM}_i^{(-\ell)}(m) - \sum_{m>T} \text{PM}_i^{(-\ell)}(m). \end{aligned} \quad (\text{A.5})$$

Thus all required quantities p_i and $A_{i\ell}$ are computed from the neighbour-restricted PMFs obtained by repeated two-point convolutions; no enumeration of $2^{|\mathcal{N}(i)|}$ configurations is needed.

Appendix B

Linearization of Update map

Inheriting from 3.22, we got

$$\begin{aligned} \begin{pmatrix} e_i(t+1) \\ r_i(t+1) \end{pmatrix} &= \begin{pmatrix} e_i^*(t+1) + \delta e_i(t+1) \\ r_i^*(t+1) + \delta r_i(t+1) \end{pmatrix} = F \begin{pmatrix} e_i(t) \\ r_i(t) \end{pmatrix} \\ &= \begin{pmatrix} (\tilde{r}_1 + p(\mathbf{e}^* + \delta \mathbf{e}(t))) \cdot (1 - [\delta e_i(t) + e_i^*] - [\delta r_i(t) + r_i^*]) \\ [\delta e_i(t) + e_i^*] + (1 - \tilde{r}_2) [\delta r_i(t) + r_i^*] \end{pmatrix}. \end{aligned} \quad (\text{B.1})$$

Noting that we can Taylor-expand $p(\mathbf{e}^* + \delta \mathbf{e}(t))$ as

$$p_i(\mathbf{e}^* + \delta \mathbf{e}(t)) = p_i(\mathbf{e}^*) + \sum_j \frac{\partial p_i}{\partial e_j} \delta e_j \quad (\text{B.2})$$

and reorganizing the terms we have

$$\begin{aligned} &\left[(\tilde{r}_1 + p_i(\mathbf{e}^*)) + \sum_j \frac{\partial p_i}{\partial e_j} \delta e_j \right] \left[(1 - e_i^* - r_i^*) - (\delta e_i(t) + \delta r_i(t)) \right] \\ &= (\tilde{r}_1 + p_i(\mathbf{e}^*)) \cdot (1 - e_i^* - r_i^*) + (1 - e_i^* - r_i^*) \cdot \sum_j \frac{\partial p_i}{\partial e_j} \delta e_j \\ &\quad - (\tilde{r}_1 + p_i(\mathbf{e}^*)) \cdot (\delta e_i(t) + \delta r_i(t)) - (\delta e_i(t) + \delta r_i(t)) \cdot \sum_j \frac{\partial p_i}{\partial e_j} \delta e_j \\ &= e_i^* + \delta e_i(t+1) \end{aligned} \quad (\text{B.3})$$

and

$$\begin{aligned} &(\delta e_i(t) + e_i^*) + (1 - \tilde{r}_2) \cdot (\delta r_i(t) + r_i^*) = \\ &(e_i^* + (1 - \tilde{r}_2) r_i^*) + (\delta e_i(t) + (1 - \tilde{r}_2) \delta r_i(t)) = r_i^* + \delta r_i(t+1) \end{aligned} \quad (\text{B.4})$$

Notice that last term in B.3 contains second order terms δ^2 which is very small when small perturbations around steady-state point is considered, and hence it's neglected in the linear approximation. Further, noticing that $q_i^* = 1 - r_i^* - e_i^*$ and using the steady-state form of 3.6 and 3.8, first terms in the right-hand side cancel with the first terms on

the left-hand side of equations B.3 and B.4. Finally, recognizing $\frac{\partial p_i}{\partial e_j}$ as A_{ij} , we are left with

$$\delta e_i(t+1) = q_i^* \cdot \sum_j A_{ij} \delta e_j - \left(\tilde{r}_1 + p_i(\mathbf{e}^*) \right) \cdot \left(\delta e_i(t) + \delta r_i(t) \right) \quad (\text{B.5})$$

$$\delta r_i(t+1) = \delta e_i(t) + (1 - \tilde{r}_2) \delta r_i(t) \quad (\text{B.6})$$

Appendix C

Global Network Metrics of Outlier Patients

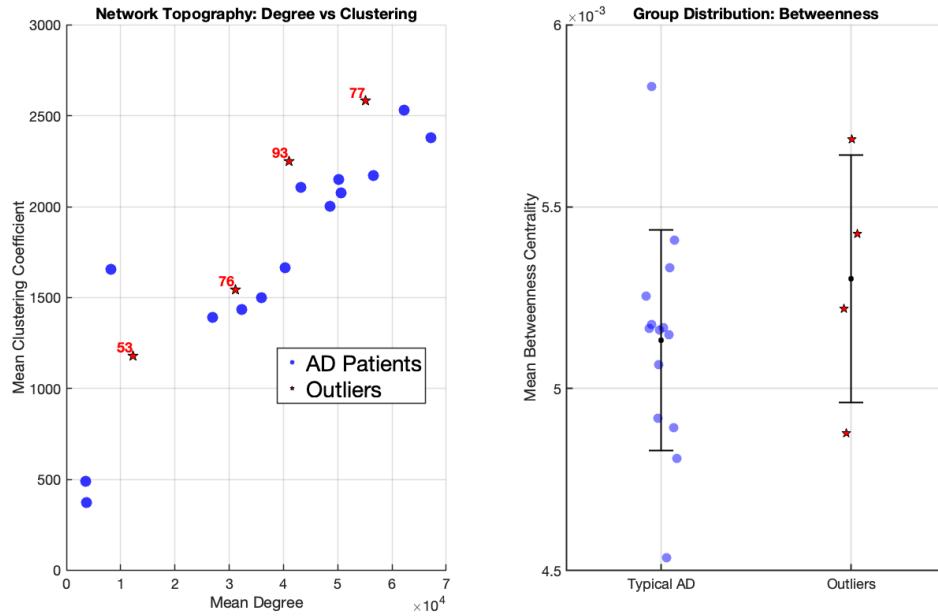


Figure C.1: Global degree, clustering coefficient and betweenness centrality of 4 outlier patients mentioned in 4.5

Global network metrics, global degree, clustering coefficient and betweenness centrality, of AD patients is plotted on the above figure and no significant difference (possibly indicating major mistake in preprocessing steps) between outlier 4.5 and other patients is observed.

Appendix D

Sub-network Similarity Analysis

In addition to whole-brain analysis, we investigated the structural-functional coupling within 7 canonical resting-state networks (Yeo parcellation). The rationale is that Alzheimer’s Disease pathophysiology may disproportionately affect specific subsystems (e.g., the Default Mode or Limbic networks) before causing global collapse. By generating simulated Functional Connectivity (FC_{sim}) at the critical threshold T_c and comparing it to empirical Functional Connectivity (FC_{real}), we aimed to identify which sub-networks exhibit the strongest divergence between simulated criticality and real-world observation. A lower similarity in specific networks would suggest that the underlying structural integrity in those regions is sufficiently degraded such that the standard critical model no longer accurately predicts their functional dynamics.

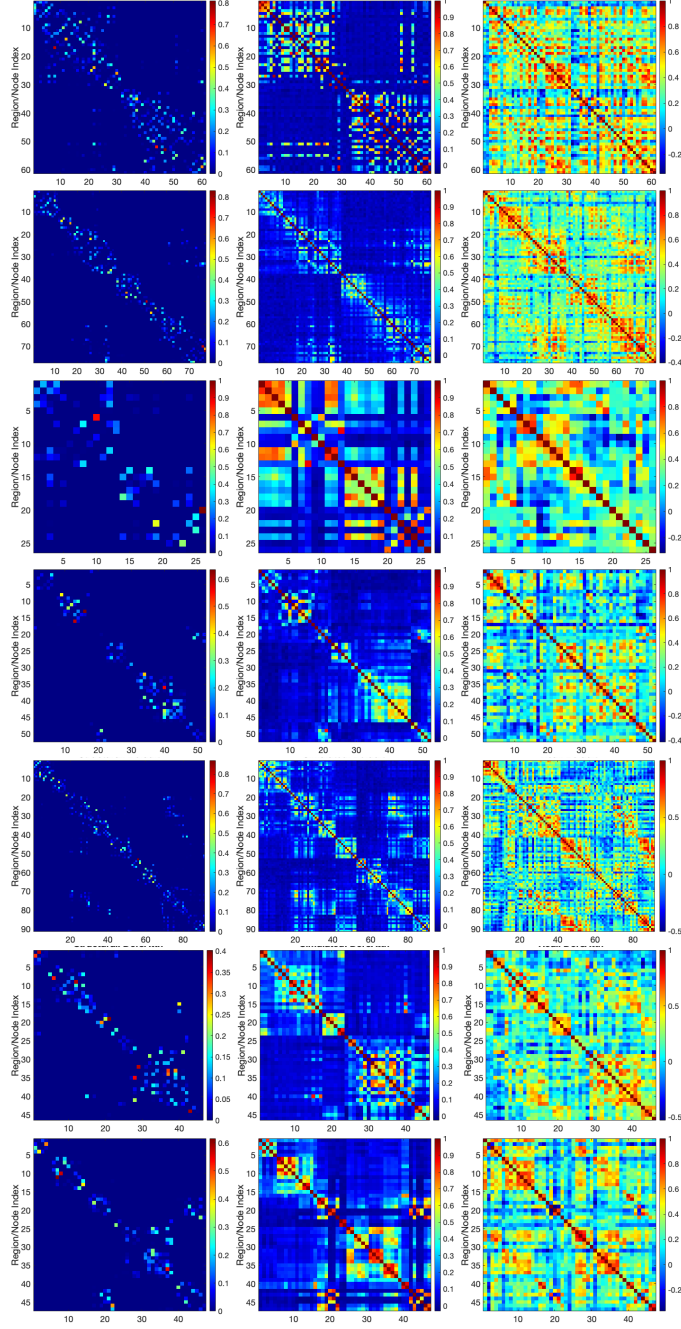


Figure D.1: Sub-network similarity analysis for 7 canonical brain networks (listed top to bottom: Visual, Somatomotor, Limbic, Control, Default, Dorsal Attention, and Salience/Ventral Attention). For each row, the matrices are: **(Left)** Structural Connectivity (SC), **(Middle)** Simulated Functional Connectivity (FC_{sim}) at T_c , and **(Right)** Empirical Functional Connectivity (FC_{real}). The comparison highlights that while some networks (e.g., Visual) maintain high fidelity between model and reality, others susceptible to AD (e.g., Limbic, Default) show distinct patterns of decoupling.

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